Permutation
&
Combination

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Permutation & Combination

Fundamental principle of counting :

Fundamental principle of counting is a method that is used to find the total number of ways of completing a work (this work consists two or more number of independent sub works). According to this principle, the total number of ways to complete the work will be equal to the product of the number of ways of completing the sub works.

So if one sub work can be done in 'm' ways following which another sub work can be done in 'n' ways, then total number of ways to complete the work is $m \times n$.

Generalization

A work consists of n independent (m_1, m_2, ..., m_n) sub works then total number of ways to complete the work is $m_1 \times m_2 \times ... \times m_n$

FPC (Fundamental Principle of Counting) is used to count some event without actually counting them.

Let us take help of some model

Model-I :

Abhishek is travelling from Ahmedabad (T_1) to Calcutta (T_3) via Bombay (T_2). There are 3 ways to travel from T_1 to T_2 and 2 ways to travel from T_2 to T_3. Then

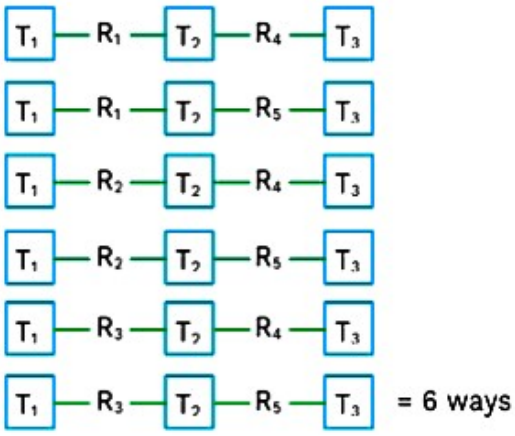
Q1. In how many ways he can travel from T_1 to T_3

Sol. He can travel from T_1 to T_2 in 3 ways

T_2 to T_3 in 2 ways

then total number of ways to travel from T_1 to T_3 is

$3 \times 2 = 6$ ways (Using FPC)



Q2. Find number of round trips ?

Sol. $T_1 \to T_2 \to T_3 \to T_2 \to T_1$

$3 \times 2 \times 2 \times 3 = 36$ ways

Q3. Number of round trips if any path can't be used twice?

Sol. $3 \times 2 \times 1 \times 2 = 12$

$T_1 \xrightarrow{3} T_2 \xrightarrow{2} T_3 \xrightarrow{1} T_2 \xrightarrow{2} T_1$

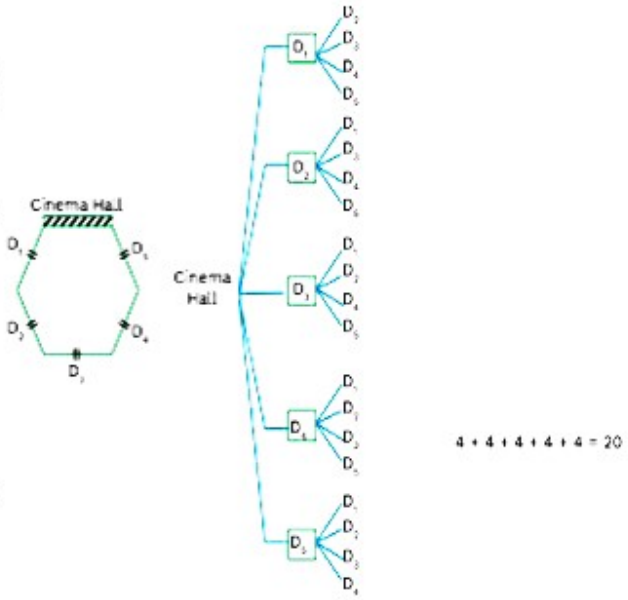
Model-II :

Mukesh wants to see a movie. So he goes to the cinema hall. There are 5 ways to enter and exit the cinema hall. Then

Q.1 Number of ways he can enter and exit through any door.

Sol. 5 ways to enter and 5 ways to exit

So $5 \times 5 = 25$ ways (Using FPC)



Q.2 Number of ways to enter and exit, if he can't exit through the same door he enters?

Sol. 5 ways to enter and 4 ways to exit

So $5 \times 4 = 20$ ways (Using FPC)

Q. **Using the digits 1, 2, 3, 4, 5, how many four digit numbers can be formed**
(i) If repetition is not allowed
(ii) If repetition is not allowed, then how many 4 digit even numbers.

Sol.

(i)

5 ways		3 ways	
↑	4 ways ↓	↑	2 ways ↓
_	_	_	_

,it means $5 \times 4 \times 3 \times 2 = 120$ numbers

(ii)

2 ways		4 ways		
↓		↓		
—	—	—	—	, it means 2 × 3 × 4 × 2= 48 numbers
	↑		↑	
	3 ways		2 ways	

Q. **If the number of five digit numbers with distinct digits and 2 at the 10th place is 336 k, then k is equal to**

(A) 8 **(B) 7** **(C) 6** **(D) 4**

Sol. Number of 5 digit numbers with distinct digits and 2 at the 10th place

= 8 × 8 × 7 × 6 = 2688 = 336 × 8.

k = 8

Q. **The number of numbers greater than 1000, but not greater than 4000, that can be formed with the digits 0, 1, 2, 3, 4, if repetition of digits being allowed, is**

(A) 372 **(B) 374** **(C) 375** **(D) 376**

Sol. The first digit from the left is 1, 2 or 3.

Each of the next 3 digits can be any one from, 0, 1, 2, 3, 4.

The number of numbers is 3 × 5 × 5× 5 = 375. But these will include the number 1000. Excluding 1000 and including 4000, the desired number is 375

Q. **The number of positive integers, which can be formed by using any number of digits 0, 1, 2, 3, 4, 5 and using each digit not more than once in each number, is**

(A) 1600 **(B) 1620** **(C) 1630** **(D) 1720**

Sol. Number of 1 digit numbers is 5 Number of 2 digit numbers is $5 \cdot 5 = 25$

Number of 3 digit numbers is $5 \cdot 5 \cdot 4 = 100$

Number of 4 digit numbers is $5 \cdot 5 \cdot 4 \cdot 3 = 300$

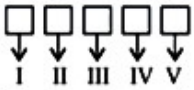
Number of 5 digit numbers is $5 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 600$

Number of 6 digit numbers is $5 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 600$

The desired number = 5 + 25 + 100 + 300 + 600 + 600 = 1630

Q. **The number of positive integers less than 40,000 that can be formed by using all the digits 1, 2, 3, 4 and 5 is equal to**

Sol.



Place I can be filled by 3 digits 1, 2, 3 otherwise number will be greater than 40,000. Place II can be filled by 4 digits (⊖ Repetition is not allowed)

Place III can be filled by 3 digits, place IV, V can be filled by 2 and 1 digits respectively. Hence, number of ways = 3 × 4 × 3 × 2 × 1 = 72 ways

Q. **Number of unsuccessful attempts to open the letter lock consists of 3 rings each marked with 10 different letters. ?**

Sol.

Total number of attempts = 10 × 10 × 10 = 10^3 = 1000
But out of these 1000 attempt, only one attempt is successful.
Required number of unsuccessful attempt = 1000 – 1 = 999

Q. **Using the digits 0, 1, 2, 3, 4, 5 (if repetition is not allowed)**
(i) How many 4 digit numbers can be formed?
(ii) How many 4 digit even number can be formed ?

Sol.

(i)

—	—	—	—
5	5	4	3
ways	ways	ways	ways

Total ways = 5 × 5 × 4 × 3 = 300 ways

(ii) Method-1

(a) When 0 is included and at the 0th place

—	—	—	—
5	4	3	1
ways	ways	ways	ways

Total ways = 5 × 4 × 3 = 60 numbers

(b) When 0 is included but not at the 0th place

—	—	—	—
4	4	3	2
ways	ways	ways	ways

Total ways = 4 × 4 × 3 × 2 = 96 numbers

Total ways = 96 + 60 = 156 numbers

Method-2

Required ways = Total – when 0 is at 1000th place

= 5 × 4 × 3 × 3 – 4 × 3 × 2

= 180 – 24 = 156 numbers

Q. **How many integers greater than 5000 can be formed with the digit 7, 6, 5, 4, and 3 using each digit at most once.**

Sol. For 4 digits number ⇒ First position can be filled by 7, 6 or 5 (that is in three ways). Hence
4 digit number = 3 × 4 × 3 × 2 = 72
5 digit number = 5 × 4 × 3 × 2 × 1 = 120
Total numbers = 192

Q. **How many natural numbers less than 30000 can be made from the digits 0, 1, 2, 3, 4, 5, 6.**

Sol. Let a five digit number be denoted by

a	b	c	d	e

Each of the places can be filled by either of 0, 1, 2, 3, 4, 5, 6 in 7 ways.

The place marked by "a" can be filled by the digits 0, 1 or 2 (since number is to be less than 30000)

Hence number of integers $= 3 \times 7 \times 7 \times 7 \times 7 = 3 \times 7^4$

In these numbers one case includes "00000" which is not a natural number.

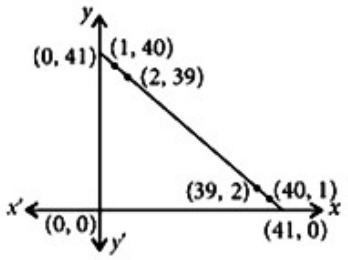
Hence number of natural numbers $= 3 \times 7^4 - 1$

Q. **The number of points, having both coordinates as integers, that lie in the interior of the triangle with vertices (0, 0), (0, 41) and (41, 0), is**

(A) 820 **(B) 780** **(C) 901** **(D) 861**

Sol. **(B)**

According to given information, we have



Now, let us count the number of points on line $x = n$, $1 \leq n < 40$, that lie in the interior of the triangle. On line $x = 40$, we have 0 point.

$x = 39$, we have 1 point

.....

$x = n$, we have $(40 - n)$ points

The total number of points $= 1 + 2 + 3 + \ldots + 39$

$$\frac{1}{2} \times 39 \times 40 = 39 \times 20 = 780$$

Q. **Number of different words that can be formed using all the letters of the words "MIRACLE".**

(a) When even position are occupied by vowels.

(b) When odd positions occupied by vowels.

Sol. **(a)** Even position ⇒

1	2	3	4	5	6	7
	×		×		×	

vowels → I, E, A

consonants → M, R, C, L

Three vowels at three position ⇒ $3 \times 2 \times 1 = 6$

Four consonants at four position ⇒ $4 \times 3 \times 2 \times 1 = 24$

Total number of ways = $6 \times 24 = 144$.

(b)

1	2	3	4	5	6	7
×		×		×		×

1st vowel have 4 choice

2nd vowel have 3 choice

3rd vowel have 2 choice

Thus total ways = $(4 \times 3 \times 2) \times (4 \times 3 \times 2 \times 1) = 576$

Q. **In how many ways 6 letters be posted in 5 different letter boxes ?**

(A) 5^6 **(B) 6^5** **(C) 5!** **(D) 6!**

Sol. **(A)**

Since, each letter can be posted in any one of the five different letter boxes. So, a letter can be posted in 5 ways. Since, there are six letters and each can be posted in 5 ways. So, total number of ways

$= 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^6$

Q. **Find the number of ways in which we can post 5 letters in 10 letter boxes.**

Sol. 1st letter ⇒ 10 boxes

2nd letter ⇒ 10 boxes

3rd letter ⇒ 10 boxes

4th letter ⇒ 10 boxes

5th letter ⇒ 10 boxes

Total ways = 10^5

Q. **Find the maximum number of car plates that can have 3 or 4 digits number plate and 0 is not included in any of the number plate?**

Sol. Here repetition of digits is allowed.

Also, numbers are formed with the digit 1, 2, 3, ..., 9

Case-I: When car plate numbers contain 3 digit, number of places to be filled up r=3. Out of the 9 digit first place can be filled by 9 ways.

Similarly, 2nd and 3rd place can be filled in 9 ways respectively.

So, when car plate number contains 3 digit, maximum number of cars = 9^3.

Case-II: When car plate number contains 4 digit, in this case number of cars to be filled up r = 4.

1st place can be filled in 9 ways.

2nd place can be filled by 9 ways and so on.

Maximum number of cars that can be numbered.= $9^3 + 9^4 = 7290$

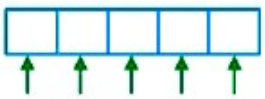
Lexicography illustrations

Know the facts

Lexicography is called science of making words.

Q. **Find total number of 5 letter word that can be formed from letters of word "TOUGH"**

Sol.



$5 \times 4 \times 3 \times 2 \times 1 = 120$

Q. **If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in a dictionary, then the position of the word SMALL is**

(A) 46th **(B) 59th** **(C) 52nd** **(D) 58th**

Sol. **(D)**

Total number of words, which are formed by using all the letters of the word SMALL =5!/2! = 60 Let's count backwards. The 60th word is SMLLA

The 59th word is SMLAL

$\therefore$ 58th word is SMALL

Q. **All the words that can be formed using alphabets A, H, L, U, R are written as in a dictionary (no alphabet is repeated). Then the rank of the word RAHUL is**
(A) 70 (B) 71 (C) 72 (D) 74

Sol. **(D)**

Since alphabets are in order A, H, L, R, U as in dictionary.
If word starts from A then number of words formed = 4!
If it starts from H then no. of words formed = 4!
If it starts from L then no. of words formed = 4!
First word starting from R with alphabet A, H, L, R, U is RAHLU. Next word is RAHUL
$\therefore$ Rank of the word RAHUL is
$4! + 4! + 4! + 1 + 1 = 3 \times 24 + 2 = 72 + 2 = 74$

Q. **If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is**

Sol. **309**

Total words starting from E = 5! = 120

Total words starting from H = 5! = 120

Total words starting from ME = 4! = 24

Total words starting from MH = 4! = 24

Total words starting from MOE = 3! = 6

Total words starting from MOH = 3! = 6

Total words starting from MOR = 3! = 6

Total words starting from MOTE = 2! = 2 and next word is MOTHER

∴ Position of the word 'MOTHER' is 120 + 120 + 24 + 24 + 6 + 6 + 6 + 2 + 1

= 240 + 48 + 18 + 3 = 309

Q.

(i) Find the number of four letter word that can be formed from the letters of the word HISTORY.
(each letter to be used at most once)

(ii) How many of them contain only consonants?

(iii) How many of them begin and end in a consonant?

(iv) How many of them begin with a vowel?

(v) How many contain the letters Y?

(vi) How many begin with T and end in a vowel?

(vii) How many begin with T and also contain S?

(viii) How many contain both vowels?

Sol.

(i) | 7 | 6 | 5 | 4 | = 840

(ii) H, S, T, R, Y

| 5 | 4 | 3 | 2 | = 120

(iii) | 5 | 5 | 4 | 4 | = 400

(iv) | 2 | 6 | 5 | 4 | = 240

(v) | | | | |

Total places available for Y is 4 other three places can be filled by

6 × 5 × 4 ways

Total ways = 480 ways

(vi) | 1 | 5 | 4 | 2 | = 40

(vii) | T | | | |

Total placed available for S = 3

Other two places can be filled by 5 × 4 = 20 ways

Total ways = 60 ways

(viii) | | | | |

For first vowel 4 places available

For second vowel 3 places available

Other two places can be filled by 5 × 4 = 20 ways

Total ways = 240 ways

NOTATION OF FACTORIAL

Notation of factorial and its Algebra :

Definition : n factorial (represented by n! or $\lfloor n$) is the product of first n natural numbers. This can also be defined as

Number of ways of arrangement of n different digits at n different places.

Note : (i) 0! = 1

1! = 1

2! = 2 × 1 = 2

3! = 3 × 2 × 1 = 6

4! = 4 × 3 × 2 × 1 = 24

5! = 5 × 4 × 3 × 2 × 1 = 120

6! = 6 × 5 × 4 × 3 × 2 × 1 = 720

(ii)Factorial of negative is not possible because we are considering the product of natural numbers.

Q. **The sum $\sum_{r=1}^{10}\left(r^2+1\right)\times(r!)$ is equal to**

(A) 11 × (11!) **(B) 10 × (11!)** **(C) (11!)** **(D) 11 × (10!)**

Sol. We have $T_r = (r^2 + 1 + r - r)r! = (r^2 + r)r! - (r - 1)r!$

$T_r = r(r+1)! - (r - 1)r!$

$\Rightarrow T_r = r(r + 1)! - (r - 1)r!$

$\therefore T_1 = 1(2!) - 0,$

$T_2 = 2(3!) - 1 \cdot 2!$

$T_3 = 3(4!) - 2 \cdot 3!$

...

...

...

$T_{10} = 10(11!) - 9(10)$

$$\sum_{r=1}^{10}\left(r^2+1\right)r! = 10(11!)$$

Q. **The sum** $S = \frac{1}{9!} + \frac{1}{3!7!} + \frac{1}{5!5!} + \frac{1}{7!3!} + \frac{1}{9!}$ **is equal to**

(A) $\frac{2^{10}}{8!}$ **(B)** $\frac{2^9}{10!}$ **(C)** $\frac{2^9}{10!}$ **(D)** $\frac{2^7}{10!}$

Sol. We have

$$S = \frac{1}{9!} + \frac{1}{3!7!} + \frac{1}{5!5!} + \frac{1}{7!3!} + \frac{1}{9!}$$

$$= \frac{1}{9!}\left[\frac{2 \times 120 + 2(72)(20) + (3024)}{120}\right] = \frac{1 \times 6144}{10! \times 12} = \frac{2^9}{10!}$$

- $(2n!) = 2^n n! [1 \cdot 3 \cdot 5 \ldots \cdot (2n - 1)]$

  Proof :

  $(2n)! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \ldots (2n - 1)(2n)$

  Take 2 common each n even terms

  $= 2^n(1 \cdot 3 \cdot 5 \ldots (2n - 1))(1 \cdot 2 \cdot 3 \ldots n)$

  $= 2^n (n!)(1 \cdot 3 \cdot 5 \ldots \cdot (2n - 1)$

Q. **The value of 2n {1. 3. 5.... (2n – 3)((2n – 1)} is**

(A) $(2n)!/n!$ **(B)** $(2n)!/2n$

(C) $n!/(2n)!$ **(D)** **None of these**

Sol. **(A)**

$$1.3.5..... (2n - 1)2^n = \frac{12345 \cdot 6 \ldots\ldots (2n - 1)(2n)2^n}{2 \cdot 4.6 \ldots 2n} = \frac{(2n)!\,2^n}{2^n(12.3 \ldots n)} = \frac{(2n)!}{n!}$$

Q. **The sum $1 \times 1! + 2 \times 2! + \ldots + 50 \times 50!$ equals**

(A) $51!$ **(B)** $51! - 1$ **(C)** $51! + 1$ **(D)** $2 \times 51!$

Sol. $1(1!) + 2(2!) + ... + 50\,(50!)$

$= (2 - 1)\,(1!) + (3 - 1)\,(2!) + (4 - 1)\,(3!) + ... + (51 - 1)(50!)$

$= (2! - 1!) + (3! - 2!) + (4! - 3!) + ... + (51! - 50!)$

$= 51! - 1! = 51! - 1$

Note :

The power of prime p in n! can be find out using the formula

$$E_p(n!) = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + ... + \left[\frac{n}{p^s}\right]$$

where s is the largest positive integer such that $p^s \le n < p^{s+1}$

Q. **Find exponent of prime 2 in (100!)**

Sol. $(2 \cdot 50)! = 2^{(50)} \cdot 50!\,(1 \cdot 3 \cdot 5 \ldots 99) \rightarrow 50$ 2's

$(50!) = (2 \cdot 25)! = 2^{(25)} \cdot 25!\,(1 \cdot 3 \cdot 5 \ldots 49) \rightarrow 25$ 2's

$(24!) = (2 \cdot 12)! = 2^{(12)} \cdot 12!\,(1 \cdot 3 \cdot 5 \ldots 23) \rightarrow 12$ 2's

$(12!) = (2 \cdot 6)! = 2^{(6)} \cdot 6!\,(1 \cdot 3 \cdot 5 \ldots 11) \rightarrow 6$ 2's

$(6!) = (2 \cdot 3)! = 2^{(3)}\,3!\,(1 \cdot 3 \cdot 5) \rightarrow 3$ 2's

$(3!) = 3 \cdot 2 \cdot 1 = 2^{(1)} \cdot (1 \cdot 3) \rightarrow 1$ 2's

$50 + 25 + 12 + 6 + 3 + 1 = 97$

Alternative Method :

$$E_2(100!) = \left[\frac{100}{2}\right] + \left[\frac{100}{2^2}\right] + \left[\frac{100}{2^3}\right] + \left[\frac{100}{2^4}\right] + \left[\frac{100}{2^5}\right] + \left[\frac{100}{2^6}\right]$$

$= 50 + 25 + 12 + 6 + 3 + 1 = 97$

Q. **25! is equal to**

(A) $2^{22}.3^{10}.5^6.7^3.11^2.13.17.19.23$

(B) $2^{10}.3^{22}.5^6.7^3.11^2.13.17.19.23$

(C) $2^{21}.3^{10}.5^6.7^3.11^2.13.17.19.23$

(D) None of these

Sol. **(A)**

Here, $\left[\frac{25}{2}\right]+\left[\frac{25}{2^2}\right]+\left[\frac{25}{2^3}\right]+\left[\frac{25}{2^4}\right]$

$= 12 + 6 + 3 + 1 = 22$

$\left[\frac{25}{3}\right]+\left[\frac{25}{3^2}\right] = 8 + 2 = 10$

$\left[\frac{25}{5}\right]+\left[\frac{25}{5^2}\right] = 5 + 1 = 6$

$\left[\frac{25}{7}\right] = 3, \left[\frac{25}{11}\right] = 2$

$25! = 2^{22} \cdot 3^{10} \cdot 5^6 \cdot 7^3 \cdot 11^2 \cdot 13 \cdot 17 \cdot 19 \cdot 23$

Q. **The number of zeros at the end of 100! is**

(A) 25 **(B) 23** **(C) 24** **(D) 26**

Sol. **(C)**

Here, $\left[\frac{100}{2}\right]+\left[\frac{100}{2^2}\right]+\left[\frac{100}{2^3}\right]+\left[\frac{100}{2^4}\right]+\left[\frac{100}{2^5}\right]+\left[\frac{100}{2^6}\right]$

$= 50 + 25 + 12 + 6 + 3 + 1 = 97$

and $\left[\frac{100}{5}\right]+\left[\frac{100}{5^2}\right] = 20 + 4 = 24$

$100! = 2^{97} \cdot 5^{24} \ldots = 2^{73} \cdot 2^{24} \cdot 5^{24} \ldots$

$= 2^{73} \cdot (2 \times 5)^{24} = 2^{73} \cdot 10^{24} \ldots$

Thus, 100! is divisible by 10^{24}.

100! ends with 24 zeros

Q. **The number of zeros at the end of 2017! is**

(A) 500 (B) 501 (C) 502 (D) 503

Sol. **(C)**

Here, $\left[\frac{2017}{5}\right]+\left[\frac{2017}{5^2}\right]+\left[\frac{2017}{5^3}\right]+\left[\frac{2017}{5^4}\right]$

$= 403 + 80 + 16 + 3 = 502$

2017! ends with 502 zeros

Q. **The number of zeros at the end of 2007! is**

(A) 499 (B) 500 (C) 501 (D) 502

Sol. **(B)**

$\left[\frac{2007}{5}\right] = 401, \left[\frac{2007}{5^2}\right] = 80, \left[\frac{2007}{5^3}\right] = 16, \left[\frac{2007}{5^4}\right] = 3$

$2007! = (3 + 16 + 80 + 401) \cdot 2\alpha\ 3\beta \ldots$

$= 5^{500} \cdot 2\alpha \cdot 3\beta \ldots$

2007! ends with 500 zeros

Q. **Find the exponent of 80 in 200!**

Sol. $80 = 2^4 \times 5$

To find the exponent of 80 in 200!, we find the exponent of 2 and 5.

Exponent of 2 is

$$\left[\frac{200}{2}\right]+\left[\frac{200}{2^2}\right]+\left[\frac{200}{2^3}\right]+\left[\frac{200}{2^4}\right]+\left[\frac{200}{2^5}\right]+\left[\frac{200}{2^6}\right]+\left[\frac{200}{2^7}\right]$$

$= 100 + 50 + 25 + 12 + 6 + 3 + 1 = 197$

Exponent of 5 is

$$\left[\frac{200}{5}\right]+\left[\frac{200}{5^2}\right]+\left[\frac{200}{5^3}\right] = 40 + 8 + 1 = 49$$

Now, exponent of 16 in 200! is $[197/4] = 49$.

Hence, exponent of 80 is 49.

Q. **Find the exponent of 11 in $^{200}C_{125}$.**

(A) 19 **(B) 12** **(C) 6** **(D) 1**

Sol. **(D)**

$$^{200}C_{125} = \frac{200!}{125!75!}$$

$\therefore$ Exponent of 11 in 200!

$$= \left[\frac{200}{11}\right] + \left[\frac{200}{11^2}\right] = 18 + 1 = 19$$

Exponent of 11 in 125!

$$= \left[\frac{125}{11}\right] + \left[\frac{125}{11^2}\right] = 11 + 1 = 12$$

$$E_{11}(75!) = \left[\frac{75}{11}\right] = 6$$

$$E(^{200}C_{125}) = 19 - (12 + 6) = 1$$

Q. **Find the number of possible values of n, such that n! has exactly 20 zeros at the end?**

Sol. In any number, number of zeros is equal to exponent of 5. Number of zeros in n! = 20

$$\Rightarrow \quad 20 = \sum_{i=1}^{\infty}\left[\frac{n}{5!}\right]$$

$$\Rightarrow \quad 20 \le \sum_{i=1}^{\infty}\frac{n}{5!} \qquad (\text{as } [x] \le x)$$

$$\Rightarrow 20 \le n\left(\frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \ldots.\right) \Rightarrow 20 \le n\left(\frac{1/5}{1-(1/5)}\right)$$

$$\Rightarrow 20 \le \frac{n}{4} \Rightarrow n \ge 80$$

If n = 80, exponent of 5 is $\left[\frac{80}{5}\right] + \left[\frac{80}{25}\right] = 19$

If n = 85, exponent of 5 is $\left[\frac{85}{5}\right] + \left[\frac{85}{25}\right] = 20$

If n = 90, exponent of 5 is $\left[\frac{90}{5}\right] + \left[\frac{90}{25}\right] = 21$

$\Rightarrow n \in \{85, 86, 87, 88, 89\}$

Permutation and Combination :

Permutation :

Permutation means arrangement in a definite order of things which may be alike or different taken some or all at a time. Hence permutation refers to the situation where order of occurrence of the events is important.

Combination :

Combination/selection/collection/committee refers to the situation where order of occurrence of the event is not important. Combination is selection of one or more things out of n things which may be alike or different taken some or all at a time.

Example :

(i) Make a number plate of 3 numbers of a car using the numbers 1, 2, 3, 4, 5, 6 is arrangement and it is considered under permutation.

(ii) If we have four letters P, Q, R, and S and if we choose any 3 letters out of these 4 to form a possible triangle, this comes under combination.

Theorem related to application of Permutation and combination :

Theorem-1

Number of arrangements of n distinct things taken all at a time is called n! and is represented by $^nP_n = P(n, n) = A_n^n = n!$

Proof :

Let these are n things arranged at n places

$n \cdot (n-1) \cdot (n-2) \ldots 3 \cdot 2 \cdot 1 = n!$

Example: Number of words that can be formed using all the letters of the words 'SHAKTI' is 6! = 720

Theorem-2

Number of arrangements (permutations) of n distinct things taken r at a time is represented by ^nP_r where $0 \le r < n$. Also

$$^nP_r = P(n, r) = A_r^n = \frac{n!}{(n-r)!}$$

Proof:

Suppose we have n distinct things $t_1, t_2, \ldots t_r$ and want to place them at r places, let's say $p_1, p_2, \ldots p_r$. Then we have n choices to fill 1st place, (n - 1) choices to fill 2nd place and so on. So using FPC, we get

$n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot [n-(r-1)]$

Total ways

$$= \frac{n(n-1)(n-2)\ldots(n-r+1)(n-r)!}{(n-r)!} = \frac{n!}{(n-r)!}$$

Hence, we can say that ${}^nP_r = \frac{n!}{(n-r)!}$

Q. **If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$ then r =**

Sol. Clearly,

$${}^{56}P_{r+6} = \frac{56!}{(56-r-6)!} = \frac{56!}{(50-r)!} \text{ and}$$

$${}^{54}P_{r+3} = \frac{54!}{(51-r)!}$$

$$\frac{56!}{(50-r)!} \times \frac{(51-r)!}{54!} = 30800$$

$$51 - r = \frac{30800}{55 \times 56} = 10 \Rightarrow r = 41$$

Theorem-3

Total number of ways to select n things out of n distinct things is 1. Also total number of ways to select 0 things out of n distinct things is 1 i.e. we reject all the things.

Theorem-4

Number of selections (combinations) of n distinct things taken r at a time is represented by ${}^nC_r = C(n,r) = \binom{n}{r} = \frac{n!}{(r)!(n-r)!}$

Proof :

Number of arrangements (permutations) of 3 distinct things selected out of 10 distinct things, suppose selection of 3 things taken at a time is 'x'. Then arrangement = (x)(3!)

$$x \cdot 3! = {}^{10}P_3$$

$$x = \frac{{}^{10}P_3}{3!} = \frac{10!}{(10-3)3!}$$

We can generate this for n distinct things.

Important results

(i) ${}^{r}C_0 = 1$

(ii) ${}^{r}C_r = 1$

(iii) Selection of r things out of n distinct things or rejection of (n - r) things out of n distinct things is same, mathematically

${}^{n}C_{n-r} = {}^{n}C_r$ If ${}^{r}C_x = {}^{r}C_y \Rightarrow x + y = n$ or $x = y$

(iv) Arrangements of r distinct things out of n distinct things can also be written as ${}^{r}P_r = {}^{n}C_r \, r!$

(v) ${}^{r}C_r + {}^{r}C_{r-1} = {}^{r+1}C_{r+1}$

Examples on ${}^{n}C_r$ and ${}^{n}P_r$

Q. **There are n points in a plane, no three of which collinear, find**

(a) Number of straight lines

(b) Number of triangles

Sol. **(a)** Number of ways by which we can select any two points gives total number of straight lines = ${}^{n}C_2$.

(b) Number of ways by which we can select any 3 non collinear points, gives total number of triangles = ${}^{n}C_3$.

Q. **Choose 3, 4, 5 points other than vertices respectively on the sides AB, BC and CA of a triangle ABC. The number of triangles that can be formed using only these points as vertices is**

(A) 220 **(B) 217** **(C) 215** **(D) 205**

Sol. **(D)**

No. of triangles = ${}^{12}C_3 - {}^{3}C_3 - {}^{4}C_3 - {}^{5}C_3 = 205$

Q. **The straight lines L_1, L_2, L_3 are parallel and lie in the same plane. A total of m points are taken on L_1, n points on L_2, k points on L_3. The maximum number of triangles formed with vertices at these points are**

(A) $^{m+n+k}C_3$ **(B)** $^{m+n+k}C_3 - {}^mC_3 - {}^nC_3$

(C) $^{m+n+k}C_3 + {}^mC_3 + {}^nC_3$ **(D) None of these**

Sol. **(D)**

Total number of points = m + n + k

Total number of triangles formed by these points taking three points at a time, is $^{m+n+k}C_3$

But out of these m + n + k, m points lies on L_1 , n on L_2 and k on L_3. By joining three points on same line we do not get a D.

Hence total no. of triangles is $^{m+n+k}C_3 - {}^mC_3 - {}^nC_3 - {}^kC_3$

Q. **The value of $(2\cdot{}^1P_0 - 3\cdot{}^2P_1 + 4\cdot{}^3P_2 - \ldots$ upto 51th term$) + (1! - 2! + 3! - \ldots$ upto 51th term$)$ is equal to**

(A) 1 – 51(51)! **(B) 1 + (51)!** **(C) 1 + (52)!** **(D) 1**

Sol. **(C)**

Clearly,

$(2\cdot{}^1P_0 - 3\cdot{}^2P_1 + 4\cdot{}^3P_2 \ldots$.upto 51 terms) + (1! – 2! + 3! – 4! + upto 51 terms)

= 2! – 3! + 4! + ... – 51! + 52! + 1! – 2! + 3! + ... + 51!

= 1 + 52!

Q. **In a class of 10 students there are 3 girls. The number of ways they can be arranged in a row, so that no 2 girls are consecutive, is k · 8! where k =**

(A) 12 **(B) 24** **(C) 36** **(D) 42**

Sol. **(D)**

The 7 boys can be arranged in a row in 7! ways. There will be 6 gaps between them and one place before them and one place after them. Therefore three girls can be arranged in these 8 places in $^8P_3 = 8 \cdot 7 \cdot 6$ ways.

$\therefore$ The desired number is $7! \times 8 \cdot 7 \cdot 6 = 42 \cdot 8!$

Q. **(i) Find the number of diagonals in an n sided polygon.**
(ii) If number of diagonals of an n sided polygon are 44. find the number of sides of the polygon.

Sol. **(i)** Number of diagonals will be can be find out by subtracting the number of sides from the ways of choosing any two points. Hence total number of diagonals

$= {}^{n}C_{2} - n = \frac{n(n-3)}{2}$

(ii) $\frac{n(n-3)}{2} = 44$

$n^2 - 3n = 88$
$n^2 - 3n - 88 = 0$
$n^2 - 11n + 8n - 88 = 0$
$(n - 11)(n + 8) = 0$
$n = 11$ or $n = -8$
Hence number of sides of polygon is 11

Q. **Eight chairs are numbered from 1 to 8. Two women and 3 men wish to occupy one chair each. First the women choose the chairs marked 1 to 4 and then the men select from amongst the remaining chairs. The number of seating arrangements is**
(A) 1250 (B) 1360 (C) 1440 (d) 1560

Sol. **(C)**

The 2 women can choose the seats marked from 1 to 4 in ${}^{4}P_{2} = 12$ ways

The 3 men can choose the seats from the remaining 6 seats in ${}^{6}P_{3} = 120$ ways

The desired number = $12 \times 120 = 1440$

Q. **A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then**
(A) m + n = 68 (B) m = n = 78
(c) m = n = 68 (D) n = m – 8

Sol. **(B)**

Case I : At least six males

$m = {}^8C_6 \cdot {}^5C_5 + {}^8C_7 \cdot {}^5C_4 + {}^8C_8 \cdot {}^5C_3$

$= 28 \times 1 + 8 \times 5 + 1 \times 10 = 28 + 40 + 10 = 78$

Case II : At least 3 females

$n = {}^5C3 \cdot {}^8C_8 + {}^5C_4 \cdot {}^8C_7 + {}^5C_5 \cdot {}^8C_6$

$= 10 \times 1 + 5 \times 8 + 1 \times 28 = 10 + 40 + 28 = 78$

Hence, $m = n = 78$.

Q. **A group of students comprises of 5 boys and n girls. If the number of ways, in which a team of 3 students can randomly be selected from this group such that there is at least one boy and at least one girl in each team, is 1750, then n is equal to**

(A) 25 **(B) 28** **(C) 27** **(D) 24**

Sol. **(A)**

Since the team consists of 3 students, in which there is at least 1 boy and at least 1 girl, so

${}^5C_1 \, {}^nC_2 + {}^5C_2 \, {}^nC_1 = 1750$ [Given]

$\Rightarrow \frac{5n(n-1)}{2} + 10n = 1750$

$5n(n-1) + 20n = 3500 \Rightarrow n(n-1) + 4n = 700$

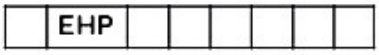
$n^2 + 3n - 700 = 0 \Rightarrow (n-25)(n+28) = 0$

$n = 25, -28$

$n = 25$ (Rejecting negative number)

Q. **How many different words can be formed from the letters of the word GANESHPURI, when the letters E, H, P are never together.**

Sol.



EHP → String

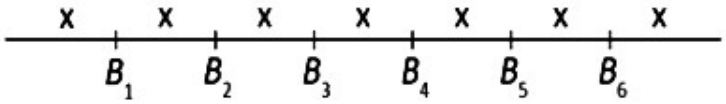
(Consider EHP as a letter or string)

So, we have to arrange one string and 7 letters Hence total number of ways = 10! - 8! × 3!
EHP can be arranged amongst themselves in 3! ways.

Q. **Six boys and six girls sit in a row. The number of ways they can be seated when the girls are separated is**

(A) $\binom{12}{6}$ **(B) (6!)2** **(C) 6! 7!** **(D) (7!)2**

Sol. **(C)**



Clearly, we have 7 vacant places for the 6 girls and so they can be seated in $7P_6 = 7!$ ways.

The desired number of ways is 6!7!

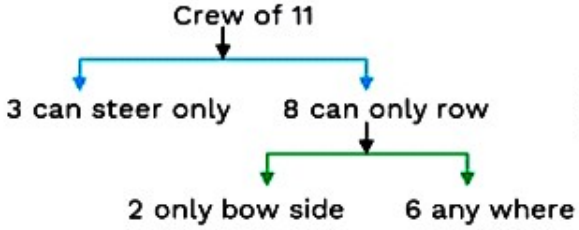
Q. **In a morse telegraphy there are 4 arms and each arm is capable of taking 5 distinct position including the position of rest. How many different signals can be transferred**

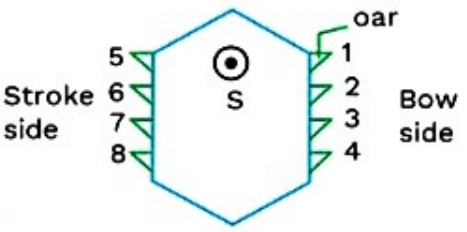
Sol. Every arm can take 5 positions.

Hence total number of signals = 5^4
There is one position in which every arm is at rest.
Hence total number of different signals = 5^4-1

Q. **An 8 oared boat to be manned from a crew of 11 of which 3 can only steer. In how many ways the staff can be arranged if 2 of the men can row only on the bow side.**

Sol.





Out of 3 person, 1 person can be selected by ${}^3C_1$ ways for steering position.

2 places can be selected out of 4 on bow side for two particular person by ${}^4C_2$ ways and they can be arranged by 2! ways. Rest six persons can be arranged by 6! ways at six places.

Thus, required ways = ${}^3C_1 \times ({}^4C_2\ (2!)) \times (6!)$

Steer — Row side — Any side

Q. **4 Boys and 4 Girls are to be seated in a line find number of ways**

(i) Number of ways they can be seated.
(ii) They can be seated so that "No two girls are together"
(iii) If not all the girls are together

or

If at least one girl is separated from rest of girls.
(iv) Boys and girls are alternate
(v) If there are 4 married couples then number of ways in which they can be seated so that each couple is together.

Sol. **(i)** They can be seatedby 8!

(ii) Out of four standing boys five gaps are there
$|\ B_1\ |\ B_2\ |\ B_3\ |\ B_4|$
Out of five select any four gaps by ${}^5C_4$ ways in which girls are arranged by
$= {}^5C_4 \times 4!$ ways.
Also standing boys are arranged by 4! ways
Required ways = $(4!) \times ({}^5C_4 \cdot 4!)$
Note : Arrangement by this method is called as gap method.

(iii) Consider $G_1\ G_2\ G_3\ G_4 \rightarrow$ as one string.

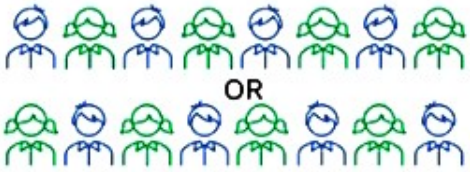
If all girls are together then total ways = $(1 + 4)!\ (4!) = (5! \times 4!)$

Four boys

String — Arrangement of four girls in string

Total number of arrangement without any restriction = 8!
Total number of ways by which not all girls are together=$8! - (5! \times 4!)$

(iv) Boys and girls are alternate then two ways of arranging respective position of boys and girls are shown below



Number of ways of arranging boys $\Rightarrow$ 4!
Number of ways of arranging girls $\Rightarrow$ 4!
Required ways = $2 \times 4! \times 4!$

(vi) Let the couples are as below

$\boxed{B_1G_1}$ $\boxed{B_2G_2}$ $\boxed{B_3G_3}$ $\boxed{B_4G_4}$

There are four string and each string has husband and wife which can be arranged in 2! ways.

Ways of arranging strings = 4!
Required ways = $(2!)^4 \times 4!$

Q. **In how many ways 10 examination papers be arranged so that the best and the worst papers never come together?**

Sol. Required ways = Total ways - When best and worst papers are together = $8 \times 9!$

Q. **A postman has to deliver 600 cards on Christmas day, the cards have been collected from the students of a class in which every student sends a card to every other student, then find the number of students under such condition?**

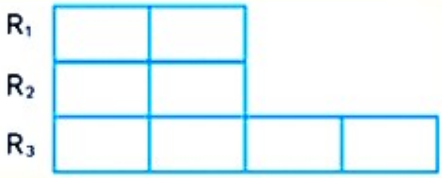
Sol. Total number of students = n
Number of pair of students = ${}^nC_2$
Two students out of n can be selected in ${}^nC_2$ ways.
Here for each pair of students, number of cards sent is 2.
If P sends card to Q, the Q also sends a card to P.
Number of cards sent = $2 \times {}^nC_2$
According to the problem $2 \times {}^nC_2 = 600$

$$\Rightarrow 2 \times \frac{n(n-1)}{2!} = 600$$

$$\Rightarrow n^2 - n - 600 = 0$$

$$\Rightarrow n = 25 \qquad [\because n \neq -24]$$

Q. **Consider the letters of the word 'PERSON', find the number of ways in which these letters can be placed in the squares of the given figure so that no row remains empty?**



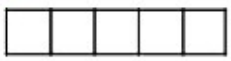
Sol. There are 6 different letters in the word 'PERSON'
Total required way = Total possible way – (1st row empty + 2nd row empty)
Total possible ways = ${}^{8}P_{6}$ (arrangement of 6 different letters in 8 boxes)
Ways when 1st row empty = 6!
(arrangement of 6 different letters in remaining 6 boxes)
Ways when 2nd row empty = 6! (arrangement of 6 different letters in remaining 6 boxes)
Total required ways = ${}^{8}P_{6}$ – (6! + 6!) = 18720

Q. **One Hindi medium school and one English medium school is under the leadership of one principal. Suppose the positions for the principal, 4 teachers of English and Hindi medium school are empty. If there are 11 candidates applied for the job, 3 out of these 11 are applied only for the job of principal and 2 only for the job in English school, then find the number of ways in which these different appointments can be filled is:**
(A) 4320 **(B) 268** **(C) 1080** **(D) 25920**

Sol. **(D)**
One sup. = ${}^{3}C_{1}$
Excepts two fixed English teachers, rest 6 teachers can be arranged at 6 positions by 6!
So Total ways = ${}^{3}C_{1} \times {}^{4}C_{2} \times 2! \times 6! = 25920$

Q. **If m denotes the number of 5-digit numbers if each successive digits are in their descending order of magnitude and n is the corresponding figure, when the digits are in their ascending order of magnitude then (m-n) has the value**
(A) ${}^{10}C_{4}$ **(B)** ${}^{9}C_{5}$ **(C)** ${}^{10}C_{3}$ **(D)** ${}^{9}C_{3}$

Sol. **(B)**



$m = {}^{10}C_{5}$

$n = {}^{9}C_5$

$m - n = {}^{10}C_5 - {}^{9}C_5 = {}^{9}C_4 = {}^{9}C_5$

Formation of groups :

(I) No. of ways in which (m + n) different things can be divided in two groups, one containing m things and other contains 'n' things is

$$^{m+n}C_n \text{ or } \frac{(m+n)!}{m!n!}$$

Q. **Out of four players P_1, P_2, P_3, P_4 form two teams one contain 3 players and other one player,**

Sol. Number of groups = $\frac{4!}{1!3!} = 4$

Note :

Actual explanation of above 4 groups
These an 4 answers so obtains

Select-3	Rejected-1
P_1, P_2, P_3	P_4
P_2, P_3, P_4	P_1
P_3, P_4, P_1	P_2
P_4, P_1, P_2	P_3

(II) If groups are equal size i.e., m = n
Total number of ways in which 2n different

things can be divided into two equal groups

$$= \frac{^{2n}C_n}{2!} = \frac{(2n)!}{(n!)(n!)(2!)}$$

We divide by 2! to avoid false counting.

Proof :

Divide P_1, P_2, P_3, P_4 in two groups

Team-A	Team-B
P_1P_2	P_3P_4
P_1P_3	P_2P_4
P_1P_4	P_2P_3
P_2P_3	P_1P_4
P_2P_4	P_1P_3
P_3P_4	P_1P_2

We see that half of the case are repeated.

Thus $\frac{4!}{2!2!}$ gives us wrong answer.

correct answer $\frac{4!}{(2!)(2!)(2!)}$

Actually, counting all such cases we observe

that regrouping appears when equal size groups are required. To avoid false counting we divide by factorial of number of equal size group's.

(III) Total numbers of ways in which (m + n + p) different things can be divided into three unequal groups m, n, p is $\frac{(m+n+p)!}{m!n!p!}$

Q. **Find the number of ways in which 21 distinct objects can be divided into three groups each containing 7 objects.**

(A) $\frac{21!}{(6!)^2}$ **(B)** $\frac{21!}{(6!)^3}$ **(C)** $\frac{(21)!}{3!(7!)^2}$ **(D)** $\frac{(21)!}{(7!)^3 3!}$

Sol. **(D)**

We know, number of ways to divide (mn) objects equally in m group

(each group will get n objects)

$$= \frac{(mn)!}{(n!)^m \times m!} = \frac{(3\times 7)!}{(7!)^3 \cdot 3!} = \frac{(21)!}{(7!)^3 \cdot 3!}$$

Important Points :

The number of ways in which 'r' group's of n different object's can be formed in such a way that 'p' groups of n_1 object, q group of n_2 object each is...

$$\text{Required ways} = \frac{n!}{(n_1!)^p (n_2!)^q (p!)(q!)}$$

$n = (n_1 + n_1 \ldots p \text{ times}) + (n_2 + n_2 \ldots q \text{ times})$

Divide by factorial of number of equal size group.

Q. **There are 3 copies each of 4 different books. The number of ways they can be arranged in a shelf is**

(A) 369600 **(B) 400400** **(C) 420600** **(D) 440720**

Sol. **(A)**

There are 4 sets of 3 identical books. The desired number is $\frac{12!}{(3!)^4} = 369600$

Q. **18 mice were placed in two experimental groups and one control group with all groups equally large. Find the number of ways in which the mice can be placed into three groups.**

(A) $\frac{18!}{(6!)^2}$ **(B)** $\frac{18!}{(6!)^3}$ **(C)** $\frac{18!}{(6!)^3 \cdot 3!}$ **(D)** $\frac{(3m)!}{(m!)^3}$

Sol. **(B)**

If 3m things are divided into 3 equal groups, then number of divisions are given by $\frac{(3m)!}{(m!)^3}$.

$\therefore$ Required ways $= \frac{18!}{(6!)^3}$

Q. **Find the number of ways in which 52 cards can be divided into 4 sets, three of them having 17 cards each and the fourth one having just one card.**

(A) $\frac{\lfloor 52}{(\lfloor 17)^3}$ **(B)** $\frac{\lfloor 52}{(\lfloor 17)^3 \lfloor 3}$ **(C)** $\frac{\lfloor 51}{(\lfloor 17)^3}$ **(D)** $\frac{\lfloor 51}{(\lfloor 17)^3 \lfloor 3}$

Sol. **(B)**

Since, 52 cards is to be divided into 3 equal sets (each containing 17 cards) and 1 set (containing only one card).

Required number of ways $= \frac{\lfloor 52}{\lfloor 17 \lfloor 17 \lfloor 17 \lfloor 3 \lfloor 1} = \frac{\lfloor 52}{(\lfloor 17)^3 \lfloor 3}$

Q. **In how many ways six different books can be distributed between four persons, so that each person gets at least one book.**

Sol. Two cases possible {1, 1, 1, 3}, {1, 1, 2, 2}

Groups $\left[\frac{6!}{(1!)^3\, 3!3!} + \frac{6!}{(1!)^2 (2!)^2\, 2!2!}\right] 4!$

Q. **Find number of ways by which five different objects given to three students, so that each person gets atleast one object.**

Sol. Two cases possible {1, 1, 3} {1, 2, 2}

$$\left[\frac{5!}{(1!)^2\,3!\times 2!}+\frac{5!}{1!(2!)^2\times 2!}\right]3!$$

Q. **Number of ways in which 8 persons can be seated in three different taxies each having 3 seats for passengers and duly numbered is**
(a) If internal arrangement of persons inside the taxi is immaterial.
(b) If internal arrangement also matters

Sol. (a) $8 \to 2, 3, 3$ $\left[\frac{8!}{2!3!3!}\times\frac{1}{2!}\right]3!$

×	×
×	D

(b) Using grouping $\left[\left(\frac{8!}{2!\,3!\,3!}\times\frac{1}{2!}\right)3!\right]3!3!3! = 9!$

or arrange 8 people in 9 seat ${}^9C_8 \times 8! = 9!$

Q. **A candidate is required to answer 6 out of 10 questions which are divided into 2 groups containing 5 questions each and he is not permitted to attempt more than 4 questions from each group. The number of ways he can make up his choice is**
(A) 120 **(B) 150** **(C) 160** **(D) 200**

Sol. **(D)**

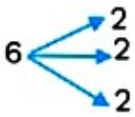
He can choose 4 from 1st group and 2 from the second group, 3 and 3 or 2 and 4.

The number of ways is $\binom{5}{4}\binom{5}{2}+\binom{5}{3}\binom{5}{3}+\binom{5}{2}\binom{5}{4}$

$= 50 + 100 + 50 = 200$

Q. **In a jeep there are 3 seat in front and three in the back, number of different ways in which six persons of different height can be seated so that every one in front is shorter than the person directly behind him,**

Sol.

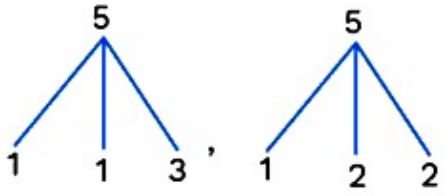


$$\frac{6! \times 3!}{(2!)^3 3!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{8 \times 6} \times 6 = 15 \times 6 = 90 \text{ Ans.}$$



Q. **(i) In how many ways can five people be divided into three groups.**
(ii) In how many ways can five people be distributed in three different rooms if no room must be empty.

Sol. **(i)**



$$\frac{5!}{(1!)^2 2! 3!} + \frac{5!}{1!(2!)^2 2!}$$

$= 10 + 15 = 25$

(ii) $25 \times 3! = 150$

Q. **(i) In how many ways can 12 different balls be divided between 2 boys, any one receiving 5 and the other 7 balls?**
(ii) In how many ways can these 12 balls be divided into groups of 5,4 and 3 balls respectively?

Sol. (i) $\frac{12!}{5!7!} \times 2!$

(ii) $\frac{12!}{3!4!5!}$

Permutation of Alike objects :

Case – I : taken all at a time
Case – II : taken some at a time

Case-I : Permutation of alike objects taken all at a time

Number of permutation of n things → p of one kind, q of another kind and r are all different } taken all at a time $= \frac{n!}{p!q!r!}$

Proof :

D A D D Y

Let the D's are different

D_1 A D_2 D_3 Y
D_1 A D_3 D_2 Y
D_2 A D_3 D_1 Y
D_2 A D_1 D_3 Y
D_3 A D_2 D_1 Y
D_3 A D_1 D_2 Y

one D A D D Y is counted as six different, word's
Let x is required ways
$x \times (6) = 6!$
$x = \frac{6!}{3!}$

Q. **The number of numbers that can be formed by using all the digits from 1, 2, 3, 4, 3, 2, 1, so that odd digits occupy odd places, is**
(A) 12 **(B) 18** **(C) 24** **(D) 36**

Sol. **(B)**

1, 1, 3, 3 are to be placed in the 4 odd places and 2, 2, 4 are to be placed in the 3 even places.

The desired number is $\frac{4!}{2!2!} \cdot \frac{3!}{2!1!} = 18$

Q. **The number of words that can be written using all the letters of the word 'IRRATIONAL' is**

(A) $\frac{10!}{(2!)^3}$ **(B)** $\frac{10!}{(2!)^2}$ **(C)** $\frac{10!}{(2!)}$ **(D) 10!**

Sol. **(A)**

There are 10 letters in the word IRRATIONAL in which there are 2'I', 2'R' and 2'A'

$\therefore$ Number of words $= \frac{10!}{2!2!2!} = \frac{10!}{(2!)^3}$

Q. **Six identical coins are arranged in a row. The total number of ways in which the number of heads is equal to the number of tails is**

(A) 9 **(B) 15** **(C) 20** **(D) 40**

Sol. **(C)**

Since coins are identical and we have to find the number of ways in which we get 3 heads and 3 tails.

Total number of ways = Number of ways in which the letters HHHTTT can be arranged $= \frac{6!}{3!3!} = 20$

Q. **Consider word ASSASSINATION, find number of ways of arranging the letters.**

(i) Number of words using all.
(ii) If no two vowels are together.
(iii) If all S are separated.
(iv) At least one S is separated from rest of the S's
(v) vowels are in the same order.
(vi) Relative position of vowels and consonant remain same.

Sol. **(i)** ASSASSINATION contains four S, three A, two N and two I.

Total ways $= \frac{13!}{(4!)(3!)(2!)(2!)}$

(ii) We have six vowels as A, A, A, I, I, O and seven consonants as S, S, S, S, N, T, N | S | S | S | S | N | T | N |

Six vowels in 8 gap's

Total ways $= {}^8C_6 \times \frac{6!}{(3!)(2!)} \times \frac{7!}{(4!)(2!)}$

(iii) | A | A | I | N | A | T | I | O | N |
Out of 10 gaps select 4

Total way = $^{10}C_4 \times \frac{9!}{(3!)(2!)(2!)}$

(iv) Total - all four S together

$$\frac{13!}{(4!)(3!)(2!)(2!)} - \frac{10!}{(3!)(2!)(2!)}$$

Consider SSSS as one string.

(v) Total ways = $\underbrace{^{13}C_6 \times 1}_{\text{arrangement of vowels}} \times \underbrace{\frac{7!}{4!2!1!}}_{\text{arrangement of consonants}}$

(vi)

v			v			v		v		v	v

Total ways = $\underbrace{\frac{6!}{3!2!}}_{\text{arrangement of vowels}} \times \underbrace{\frac{7!}{4!2!}}_{\text{arrangement of consonants}}$

Q. **How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which no two S are adjacent?**
(A) $7 \cdot {}^6C_4 \cdot {}^8C_4$ **(B)** $8 \cdot {}^6C_4 \cdot {}^7C_4$ **(C)** $6 \cdot 7 \cdot {}^8C_4$ **(D)** $6 \cdot 8 \cdot {}^7C_4$

Sol. **(A)**

Leaving S, we have 7 letters M, I, I, I, P, P, I. Number of ways of arranging them
$= \frac{7!}{2!4!} = 7 \cdot 5 \cdot 3$
and four S can be put in 8 places in $^8C_4$ ways.

$\therefore$ The required number of ways $= 7 \cdot 5 \cdot 3 \cdot {}^8C_4$

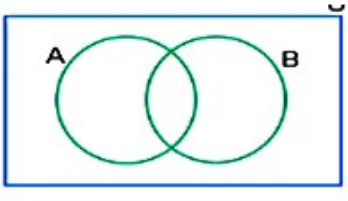
$= 7 \cdot {}^6C_4 \cdot {}^8C_4$.

Principle of inclusion and exclusion

We have been using principle of inclusion and exclusion right from Set Theory.

Consider the following Venn diagram for sets A and B.

Here U is universal set.

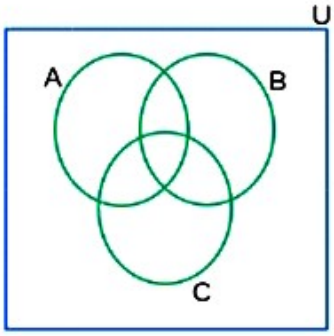


Clearly, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Here, n(A) is number of elements in set A. It is called cardinality of set A.

We are subtracting or excluding $n(A \cap B)$ from $n(A \cup B)$ as elements in set $(A \cap B)$ are counted twice in n(A) + n(B).

Now consider the following Venn diagram for sets A, B and C.



Here $n(A \cup B \cup C) = n(A) + n(B) + n(C) - [n(A \cap B) + n(B \cap C) + n(A \cap C)] + n(A \cap B \cap C)$.

Generalisation, $n(A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_n)$ =

$$\sum_{i=1}^{n} n(A_i) - \sum_{1 \le i < j \le n} n(A_i \cap A_j) + \sum_{1 \le i < j < k \le n} n(A_i \cap A_j \cap A_k) + \ldots + (-1)^{n-1} \sum n(A_1 \cap A_2 \cap \ldots \cap A_n)$$

Q. **Find the number of positive integers from 1 to 1000, which are divisible by at least one of 2, 3 or 5.**

Sol. Let A_k be the set of positive integers from 1 to 1000, which are divisible by k.
Clearly, we have to find $n(A_2 \cup A_3 \cup A_5)$.
If [.] denotes the greatest integer function, then

$$n(A_2) = \left[\frac{1000}{2}\right] = 500$$

$$n(A_3) = \left[\frac{1000}{3}\right] = 333$$

$$n(A_5) = \left[\frac{1000}{5}\right] = 200$$

Also, $$n(A_2 \cap A_3) = \left[\frac{1000}{6}\right] = 166$$

$$n(A_3 \cap A_5) = \left[\frac{1000}{15}\right] = 66$$

$$n(A_2 \cap A_5) = \left[\frac{1000}{10}\right] = 100$$

and $n(A_2 \cap A_3 \cap A_5) = \left[\frac{1000}{30}\right] = 33$

So, $n(A_2 \cup A_3 \cup A_5) = 500 + 333 + 200 - 166 - 66 - 100 + 33 = 734$

Note that number of positive integers from 1 to 1000, which are not divisible by any of 2, 3 or 5

$= n(A_2' \cap A_3' \cap A_5')$

$= n(U) - n(A \cup B \cup C)$

$= 1000 - 734 = 266$

Q. **Number of ways in which 2 African, 2 Bhartiya, 1 Canadian, 1 Dominican, 1 Ethiopian can sit on a round table so that no two alike letters are together?**

Sol. Suppose all these letters are different then total number of circular permutations is 6!

Since all the letters are different then consider 2A as A_1 and A_2

If n(A) is the number of arrangements when A_1 and A_2 are together.

$\therefore$ $n(A) = 5!\ 2!$ (suppose A_1A_2 is one unit)

If n(B) is the number of arrangements when B_1 and B_2 are together.

$\therefore$ $n(B) = 5!\ 2!$ (suppose B_1B_2 is one unit)

$\therefore$ $n(A \cup B)$ = number of arrangements in which two A's and two B's are together.

$= 4!\ 2!\ 2!$

Now we want number of arrangements in which alike letters are separated.

i.e., $n(A' \cap B') = n(U) - n(A \cup B)$

$= n(U) - [n(A) + n(B) - n(A \cap B)]$

$= 6! - [5!2! + 5!2! - 4!\ 2!\ 2!]$

$= 720 - [240 + 240 - 96]$

$= 720 - 384 = 336$

Q. **Total number of permutation of letters a, b, c, d, e, f, g taken all together if neither 'beg' nor 'cad' pattern appear.**

Sol. When there is not any restrictions,

$n(U) = 7!$

Number of permutations in which 'beg' (b e g) a c d f pattern always appear are, $n(A) = 5!$

Number of permutations in which 'cad' (c a d) b e f g pattern always appear are, n(B) = 5!

Number of permutation in which both 'beg'

(c a d) (b e g) f

and 'cad' pattern appear are, $n(A \cap B) = 3!$

Then number of permutations in which 'beg' and 'cad' pattern do not appear

$= n(A' \cap B') = n(U) - n(A \cup B)$

$= n(U) - [n(A) + n(B) - n(A \cap B)]$

$= 7! - [5! + 5! - 3!] = 4806$

Q. **How many words can be formed using all the letters of the word HONOLULU if no two alike letters are together.**

Sol. Consider three events A, B, C where A represent ways when OO together, B represents when LL together, C represents when UU is together.

Using principle of Inclusion and Exclusion

Words under the condition = Total ways - [When all three alike letters together + when 2 alike letters together + when one alike letter together]

$= \text{Total ways} - [n(E_3) + n(E_2) + n(E_1)]$ ...(i)

$$\text{Total ways} = \frac{8!}{(2!)(2!)(2!)} = 5040$$

(a) $n(E_3) = A \cap B \cap C$ (Region 7)

i.e.,

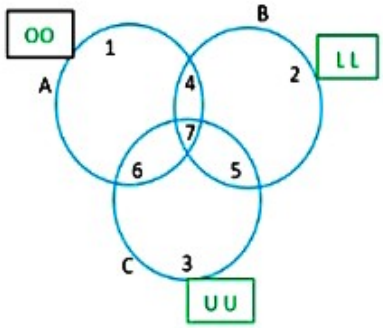
H	N	OO	LL	UU

$n(E_3) = 5! = 120$

(b) $n(E_2) = 3\left[(A \cap B) - (A \cap B \cap C)\right]$

or (Region 4+5+6)

$$3\left(\frac{6!}{2!} - 5!\right) \quad (n(E_3)) \qquad = 720$$



i.e.

H	OO	N	LL	UU

(c) $n(E_1) = 3\left[A - \{(A \cap B) + (A \cap C)\} + (A \cap B \cap C)\right]$

$$= 3\left[\frac{7!}{(2!)(2!)} - \left(\frac{6!}{2!} \times 2\right) + 5!\right] = 1980$$

Put in 1st

Required ways = 5040-[120+720+1980]=2220

Q. **2, 3, 4, 5 numbers put on the four faces of a teterahedral dice,if lowest face can be considered as the outcome, then find the number of ways in which a total of 30 can occur in 7 throws.**

Sol. 7 throws outcome whose sum is equal to 30 can be obtained in following way.

Category	Number of ways
5, 5, 5, 5, 5, 2, 3	$\frac{7!}{5!} = 42$
5, 5, 5, 5, 4, 4, 2	$\frac{7!}{(4!)(2!)} = 105$
5, 5, 5, 5, 4, 3, 3	$\frac{7!}{(4!)(2!)} = 105$
5, 5, 5, 4, 4, 4, 3	$\frac{7!}{(3!)(3!)} = 140$
5, 5, 4, 4, 4, 4, 4	$\frac{7!}{(2!)(5!)} = 21$

Total ways = 42 + 105 + 105 + 140 + 21 = 413

PERMUTATION OF THE OBJECTS TAKEN SOME AT A TIME :

Category	Process of selection	Selection	Arrangement
1. 2 alike, 4 different	**(a)** 2 alike PP, RR, OOO $\rightarrow {}^{3}C_{1}$ **(b)** 4 different from 5 possible option $\rightarrow {}^{5}C_{4}$	${}^{3}C_{1} \times {}^{5}C_{4} = 15$	$15 \times \frac{6!}{2!}$
2. 3 alike, 3 different	**(a)** 3 alike OOO $\rightarrow {}^{1}C_{1}$ **(b)** 3 different out of 5 different option $\rightarrow {}^{5}C_{3}$	${}^{1}C_{1} \times {}^{5}C_{3} = 10$	$10 \times \frac{6!}{3!}$
3. 2 alike of one kind 2 alike other kind and 2 different	**(a)** 2 alike of one and 2 alike of other PP, RR, OOO, $\rightarrow {}^{3}C_{2}$ **(b)** 2 out of 4 different $\rightarrow {}^{4}C_{2}$	${}^{3}C_{2} \times {}^{4}C_{2} = 18$	$18 \times \frac{6!}{2!2!}$
4. 3 alike of one kind +2 alike other kind +1 different	**(a)** OOO, PP, PR $\rightarrow {}^{1}C_{1} \times {}^{2}C_{1}$ **(b)** 2 out of 4 different $\rightarrow {}^{4}C_{1}$	${}^{2}C_{1} \times {}^{4}C_{1} = 8$	$8 \times \frac{6!}{3!2!}$
5. 2 alike, 2 alike + 2 alike	PP, RR, OOO $\rightarrow {}^{1}C_{1} \times {}^{1}C_{1} \times {}^{1}C_{1}$	${}^{1}C_{1} = 1$	$1 \times \frac{6!}{2!2!2!}$
6. All 6 different	6 possible option P, R, O, T, I, N $\rightarrow {}^{6}C_{6}$	$\rightarrow {}^{6}C_{6} = 1$	$6!$
Total		Number of section = 53	Number of words=11130
Then total number of words formed = 11130			

Q. **How many 5 lettered words can be formed using the letters of the word "INDEPENDENCE"**

Sol. There are E → 4, N → 3, D → 2, P → 1, I → 1, C → 1

Category	Seleection	Arrangement
4 alike and 1 diff.	$1 \times {}^5C_1 = 5$ Eg. EEEED	$\frac{5!}{4!} \times 5 = 25$
3 alike and 2 diff.	${}^2C_1 \times {}^5C_2 = 20$ EEEDD	$\frac{5!}{3!} \times 20 = 400$
3 alike and 2 alike of diff. kind	${}^2C_1 \times {}^2C_1 = 4$ EEEDD	$\frac{5!}{3! \times 2!} \times 4 = 40$
2 alike and 3 diff.	${}^3C_1 \times {}^5C_3 = 30$ EENDI	$\frac{5!}{2!} \times 30 = 1800$
2 alike + 2 other alike and 1 diff	${}^3C_2 \times 4 = 12$ EENND	$\frac{5!}{2! \times 2!} \times 12 = 360$
all five diff.	${}^6C_5 = 6$ EDIPC	$6 \times 5! = 720$

Number of arrangements = 3345

Q. **Find the number of words each consisting 5 letters from the letters of the word "M I S S S I S S I P P I"**

Sol. S → 4, I → 4, P → 2, M → 1

Category	Seleection	Arrangement
(1) 4 alike, 1 different	${}^{2}C_{1} \times {}^{3}C_{1} = 6$	$6 \times \frac{5!}{4!}$
(2) 3 alike, 2 alike	${}^{2}C_{1} \times {}^{2}C_{1} = 4$	$4 \times \frac{6!}{2!3!}$
(3) 3 alike, 2 different	${}^{2}C_{1} \times {}^{3}C_{1} = 6$	$6 \times \frac{6!}{3!}$
(4) 2 alike, 2 alike 1 different	${}^{3}C_{2} \times {}^{2}C_{1} = 6$	$6 \times \frac{5!}{2!2!}$
(5) 2 alike, 3 different	${}^{3}C_{1} \times {}^{3}C_{3} = 3$	$3 \times \frac{5!}{2!}$
Total	Number of Selection = 25	Number of Arrangement = 1350

Then total number of words formed = 1350

Q. **Determine the number of permutations of the letters of the word MATHEMATICS.**

Sol. M – 2, A – 2, T – 2, H – 1, E – 1, I – 1, C – 1, S – 1

Total permutations = $\frac{11!}{2!2!2!}$

Q. **In how many ways can be letters of the word RESTRICTION be arranged so that the vowels never occur together.**

Sol. Consonants ⇒ R – 2, S – 1, T – 2, C – 1, N – 1

Vowels $\Rightarrow$ E – 1, I – 2, O – 1
Using gap method,

Total ways = $\frac{7!}{2!2!}.{}^{8}C_{4}.\frac{4!}{2!}$

Q. **How many ways can be formed with the letters of the word RESTRICTION without changing the relative order of the vowels and consonants.**

Sol. $_\ V\ _\ _\ _\ V\ _\ _\ V\ V\ _$

Total ways = $\frac{4!}{2!}.\frac{7!}{2!2!}$

Q. **In how many ways can the letters AAABBCD be arranged so that the two B's are together but no two A's are together.**

Sol. A A A B B C D
A – 3, B – 2, C – 1, D – 1

| BB | C | D |

$3! \cdot {}^{4}C_{3} \cdot 1 = 24$

CIRCULAR PERMUTATION

When object are different

Arrangements of objects in a closed curve considered as circular permutation while arrangements of objects in a line (row) considered as linear permutation. That's why, in circular permutation, one object can be considered as fixed and the remaining objects are arranged as in the case of a linear arrangements.

Note:

(i) The number of circular permutation of **n** distinct objects is **(n – 1)!**

(ii) Taken r things out of n distinct things at a time and arranged along circle in ${}^{n}C_{r}.(r-1)!$ ways.

(iii) If anticlockwise and clockwise are considered to be same total number of circular permutation given by $\frac{(n-1)!}{2}$.

If we arrange flowers or garland beads in a neckless then there is no distinction between clockwise and anticlockwise direction.

(iv) If we have n different things taken r at a time in form of a garland or neckless
Required number of arrangements

$$= \frac{{}^{n}C_{r} \cdot (r-1)!}{2}$$

(v) The distinction between clockwise and anticlockwisle is ignored when a number of people have to be seated around a table so as not to have the same neighbours.

Q. **If there are 7 persons in a group, the number of circular permutations of 7 persons when clockwise and anticlockwise arrangements of 7 persons are same, is**
(A) 120 **(B) 240** **(C) 360** **(D) 480**

Sol. **(C)**

We know that, number of permutations of n different objects when clockwise and anticlockwise

arrangement of objects are same $= \frac{n!}{2n}$

$\therefore$ Required arrangements $= \frac{7!}{2 \times 7} = \frac{6!}{2} = \frac{720}{2} = 360$

Q. **There are 18 objects out of which two are b_1 and b_2. Find the number of ways in which we can arrange them at a round table so that there is exactly one object between the objects b_1 and b_2**
(A) 2 × 14! **(B) 2 × 15!** **(C) 2 × 16!** **(D) 2 × 17!**

Sol. **(C)**
Let a object M will be placed between the objects b_1 and b_2. Now, object M can be chosen from 16 objects (excluding b_1 and b_2) in 16 ways.
Now, consider the objects b_1, b_2 and M as one object, we have (18 – 3 + 1) = 16 objects in all. These 16 objects can be arranged around a circle in (16 – 1)! = 15! ways.
But b_1 and b_2 can be arranged themselves by 2! ways.
Hence, the total number of ways = 16 × 15! × 2! = 2 × 16!

Q. **The number of ways in which 29 persons be seated around a table if there are 9 chairs, is**

(A) $\frac{^{29}P_9}{2}$ **(B)** $\frac{\left(^{29}P_9\right)}{9}$ **(C)** $\frac{^{29}P_9}{3}$ **(D) None of these**

Sol. **(B)**

In case of circular table, clockwise and anticlockwise arrangements are different,

number of circular permutations of n different things taken r at a time $= \frac{^nP_r}{r}$

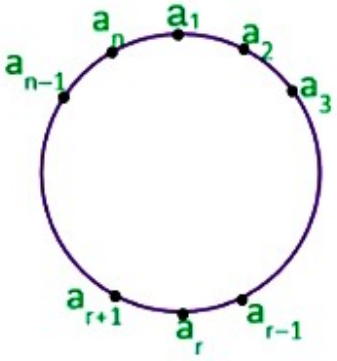
$\therefore$ Required arrangements $= \frac{\left(^{29}P_9\right)}{9}$

Q. **n different things are arranged in a circle. In how many ways can three object be selected if no two objects are consecutive**

Sol.

Select 1 thing, let's say a_1 by $^nC_1$ then we can not select a_2 and a_n. Now we are left with (n - 3) things out of which we have to select 2 things, leave those two, now we have (n - 5) things. Select two positions in between these (n - 5) by $^{(n-5+1)}C_2$

Hence $\frac{^nC_1 \times ^{(n-4)}C_2}{3}$ is the required number of ways.



Q. **Find number of circular permutation of n persons if two specific people are never together.**

Sol. Required ways = Total – when A and B are always together

$= (n-1)! - (n-2)! \times 2 = (n-2)! [n-1-2]$

$= (n-2)! (n-3)$.

Q. **The number of ways in which four boys can be seated around a round table in four chairs of different colours is**

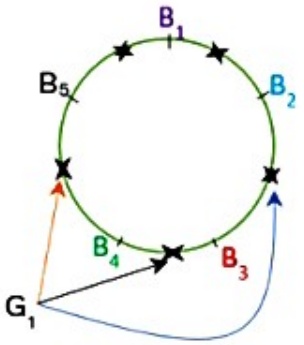
(A) 24 **(B) 12** **(C) 23** **(D) 64**

Sol. **(A)**

Required number of ways = 4! = 24

Q. **Find number of ways in which 5B and 5G can be seated on a circle alternately if a particular B_1 and G_1 are never adjacent to each other in any arrangement.**

Sol. For B_1, G_1 not to be together. G_1 must select from 3 possible gaps in ${}^3C_1$ ways



$4! \times {}^3C_1 \times 4! = 1728$

Boys G_1

Q. **In how many ways triangle can be constructed by joining vertices of a quindecagon if none of the sides of triangle can be sides of quindecagon.**

Sol. By method in above question

$$= \frac{{}^{15}C_1 \times {}^{11}C_2}{3} = 275$$

Q. **In how many ways can 5 men and 5 women be seated at a round table if**
(i) there is no restriction
(ii) all the five women sit together
(iii) no two women sit together
(iv) not more than four women sit together

Sol. **(i)** Total persons = 10

So 9!

(ii) $\boxed{W_1 W_2 W_3 W_4 W_5}\ M_1\ M_2\ M_3\ M_4\ M_5$

So (6 – 1)! 5! = 5! 5!

(iii) Using Gap method,
$4! \cdot {}^5C_5 \cdot 5! = 4!\ 5!$

(iv) Total – (all women are together) $= 9! - 5!\ 5!$

Q. **In how many ways can 5 men and 3 women be seated at a round table if**
(i) no two women sit together
(ii) two particular women must sit together, while the third one must not sit beside those two.

Sol. **(i)** Using gap method,
$(5-1)! \cdot {}^5C_3 \cdot 3! = 4! \cdot {}^5P_3$

(ii) $\boxed{W_1W_2}\,W_3$

Using gap method,
$= 4! \times {}^5C_2 \times 2! \times 2!$

Q. **Find the number of ways in which n different beads can be arranged to form a necklace.**

Sol. Clockwise and Anticlockwise distinction is ignored
So $\dfrac{(n-1)!}{2}$

TOTAL NUMBER OF COMBINATIONS OR SELECTION OR COLLECTION :

We know that $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \ldots + {}^nC_rx^r + \ldots + {}^nC_nx^n$

Now replace x by 1
then $2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n$

Case-I :

Selection of one or more things out of n things.
When all the things are different total number of selections.
One things can be selected in ${}^nC_1$ ways
Two things can be selected in ${}^nC_2$ ways
Three things can be selected in ${}^nC_3$ ways
⋮
⋮

n things can be selected in ${}^nC_n$ ways
Total ways $={}^nC_1 + {}^nC_2+{}^nC_3+...+{}^nC_n=2^n-1$

Case-II: The number of ways of selecting zero or more things out of n identical things is n + 1.

Proof :

Selecting none thing = 1 way
Selecting 1 thing = 1 ways
Selecting 2 things = 1 way
⋮
⋮
Selecting n things = 1 way
Total number of ways = 1 + 1 + ... + (n +1) times = (n + 1)

Case-III:

Number of ways of selecting one or more things out of which p are alike of one kind, q are alike of second kind, r alike of third kind, while s are different is $(p + 1)(q + 1)(r + 1)\ 2^s - 1$

Proof :

Selecting none thing (out of p alike things) = 1 way
Selecting 1 thing (out of p alike things) = 1 ways
Selecting 2 things (out of p alike things) = 1 way
⋮
⋮
Selecting p things (out of p alike things) = 1 way
Total number of ways = 1 + 1 + 1 + ... + (p + 1) times = (p + 1)
Similarly for q alike, total ways = q + 1
Similarly for r alike, total ways = r + 1
For s different things total ways of selecting will be 2^s, i.e. any item is selected or not.
So total number of required ways = $(p + 1)(q + 1)(r + 1)2^s-1$
(1 is subtracted when no item is selected)

Q. A man has 6 friends. Number of different ways he can invite 2 or more for a dinner is
(A) 56 **(B) 72** **(C) 28** **(D) 57**

Sol. **(D)**

Required number = ${}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6$

$= 15 + 20 + 15 + 6 + 1 = 57$

Q. **The question paper in the examination contains three sections A, B, C. There are 5, 4, 3 questions in each sections A, B, C respectively. A student has freedom to answer any number of questions attempting at least one from each section. In how many way can the paper be attempted by a student?**

(A) 3250 **(B) 3225** **(C) 3255** **(D) 3205**

Sol. **(C)**

As section A having 5 questions, the student can select at least one question out of 5 in $(2^5 - 1)$ ways.

Similarly, the student can select at least one question from B and C by (2^4-1) and (2^3-1) ways.

Hence, total number of ways in which a student can attempt the paper

$= (2^5 - 1)(2^4 - 1)(2^3 - 1)$

$= 31 \times 15 \times 7 = 3255$

Q. **Find the number of ways in which one can make a selection of at least one book on each of three subjects physics, chemistry and mathematics from a library containing 4 different books on physics, 6 different books on chemistry and 3 different books on mathematics?**

(A) 6605 **(B) 6610** **(C) 6615** **(D) none of these**

Sol. **(C)**

A selection of at least one book on physics can be made out of 4 different books in ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4 - 1 = 15$ ways

Similarly, atleast one book on chemistry can be selected in $2^6 - 1 = 63$ ways and at least one mathematics book out of 3 can be selected in $2^3 - 1 = 7$ ways.

Hence, the required number of ways of selecting at least one book on each of three subjects $= 15 \times 63 \times 7 = 6615$

Total number of combinations in different cases:

(a) The number of combinations of n **different things** taking some or all (or atleast one) at a time = ${}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n - 1$

(b) The number of ways to select some or all out of (p + q + r) things where p are alike of first kind, q are alike of second kind and r are alike of third kind is

$= (p + 1)(q + 1)(r + 1) - 1$

(c) The number of ways to select some or all out of **(p + q + t)** things where p are alike of first kind, q are alike of second kind and remaining t are different is

$= (p + 1)(q + 1)2^t - 1$

PROBLEMS BASED ON NUMBER THEORY

Note that every natural number except 1 has atleast 2 divisors. If it has exactly two divisors then it is called a prime. System of prime numbers begins with 2. All primes except 2 are odd. A number having more than 2 divisors is called a composite. 2 is the only even number which is not composite. A pair of natural numbers are said to be relative prime or coprime if their HCF is one. For two natural numbers to be relatively prime it is not necessary that one or both should be prime. It is possible that they both are composite but still coprime. eg. 4 and 25. Note that 1 is neither prime nor composite however it is coprime with every other natural number. A pair of primes are said to be twin if their non-negative difference is 2 e.g. 3 and 5; 5 and 7 etc.

Number of divisors and their sum :

(a) Every natural number N can always be put in the form $N = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_k^{\alpha_k}$ where $p_1, p_2, \ldots p_k$ are distinct primes and $\alpha_1, \alpha_2, \ldots, \alpha_k$ are non-negative integers.

(b) If $N = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_k^{\alpha_k}$ then the number of divisor of N is equivalent of number of ways of selecting zero or more objects from the groups of identical objects, (p_1, p_1,...,α_1 times) (p_2, p_2, ...α_2 times), (p_k, p_k, ...,α_k times)= $(\alpha_1 + 1)(\alpha_2 + 1) \ldots (\alpha_k + 1)$ which includes 1 and N also.

(c) All the divisors excluding 1 and N are called proper divisors.
Also number of divisors of N can be seen as number of different terms in the expansion of

$$\left(p_1^0 + p_1^1 + p_1^2 + \ldots + p_1^{\alpha_1}\right) \times \left(p_2^0 + p_2^1 + p_2^2 + \ldots + p_2^{\alpha_2}\right) \times \ldots \times \left(1 + p_k + p_k^2 + \ldots + p_k^{\alpha_k}\right)$$

Hence, sum of the divisors of N is

$$\left(1+p_1+p_1^2+\ldots+p_1^{\alpha_1}\right)\times\left(1+p_2+p_2^2+\ldots+p_2^{\alpha_2}\right)\times\ldots\times\left(1+p_k+p_k^2+\ldots+p_k^{\alpha_k}\right)$$

$$=\frac{p_1^{\alpha_1+1}-1}{p_1-1}\frac{p_2^{\alpha_2+1}-1}{p_2-1}\ldots\frac{p_k^{\alpha_k+1}-1}{p_k-1}$$

(d) Number of ways of putting N as a product of two natural numbers is

$$\frac{1}{2}(\alpha_1+1)(\alpha_2+1)\ldots(\alpha_k+1)$$

if N is not a perfect square.

If N is a perfect square, then this is

$$\frac{1}{2}[(\alpha_1+1)(\alpha_2+1)\ldots(\alpha_k+1)+1]$$

(e) If $N = p^a \times q^b \times r^c \ldots$ p, q are prime number and a, b are natural number.

$$I=\frac{(\text{Total number of divisors})}{2}$$

where I is the number of required ways if N is not a perfect square.

$$I=\frac{(\text{Total number of divisors}+1)}{2}$$

where I is the number of required ways if N is a perfect square.

Q. **Consider the number N = 75600**

Now answer the following question

(i) Number of divisors

(ii) Number of proper divisors

(iii) Number of odd divisor

(iv) Number of even divisor

(v) Number of divisors divisible by 5

(vi) Number of divisors divisible by 10

(vii) Sum of all divisors

(viii) Number of ways in which N can be resolved as product of two divisors

Sol.

(i) $N = 2^4 \times 3^3 \times 5^2 \times 7^1$

Number of divisor $= (4+1)(3+1)(2+1)(1+1)$

$= 5 \times 4 \times 3 \times 2 = 120$

(ii) Proper divisor will be other than 1 and number itself

So proper divisor = 120 – 2 = 118

(iii) Number of odd divisor can be obtained by choosing $3^3 \times 5^2 \times 7^1$

Which can be formed by $(3+1)(2+1) \times (1+1) = 24$ ways

(iv) It can be obtain by selecting at least one '2' in $2^4.3^3.5^2.7^1$
which can be done in $4\times(3+1)\times(2+1)\times(1+1) = 96$ ways

(v) At least one 5 must be there in $2^4 \times 3^3 \times 5^2 \times 7^1$
$(4 + 1)(3 + 1)(2)(1 + 1) = 80$

(vi) For divisibility by 10 number most be divisible by 2 and 5
So total divisor must contain at least one 2 and at least one 5 in $2^4 \times 3^3 \times 5^2 \times 7^1$
So that ways $= 4 \times (3 + 1) \times (2) \times (1 + 1) = 64$ ways

(vii) It can be obtained from

$$\left(2^0 + 2^1 + 2^2 + 2^3 + 2^4\right)\left(3^0 + 3^1 + 3^2 + 3^3\right)$$
$$\left(5^0 + 5^1 + 5^2\right)\left(7^0 + 7^1\right)$$
$$= \left(2^5 - 1\right) \times \frac{\left(3^4 - 1\right)}{2} \times \frac{\left(5^3 - 1\right)}{4} \times \frac{\left(7^2 - 1\right)}{6}$$

(viii) $N = 2^4 \times 3^3 \times 5^2 \times 7^1$
Total number of divisor $(N_T) = (4 + 1) \times (3 + 1) \times (2 + 1) \times (1 + 1) = 120$
If (N_T) is even then required number of ways $= \frac{N_T}{2} = 60$

$$= 2 \times \frac{\left(3^5 - 1\right)}{2} \times \frac{\left(5^8 - 1\right)}{4} \times \frac{\left(7^3 - 1\right)}{6} = \frac{\left(3^5 - 1\right)\left(5^8 - 1\right)\left(7^3 - 1\right)}{24}$$

Q. **The number of divisors of the form $4n + 2$, $n \geq 0$ of the integer 240 is**
(A) 3 **(B) 4** **(C) 8** **(D) 10**

Sol. **(B)**

$240 = 2^4 \cdot 3 \cdot 5.$

A divisor is of the form $4n + 2 = 2(2n + 1) = 2 \cdot$ (odd natural number)

The divisors are $2 \cdot 1, 2 \cdot 3, 2 \cdot 5, 2 \cdot 3 \cdot 5$, i.e., 2, 6, 10, 30

Q. **Find the number of odd proper divisors of $3^p \times 6^m \times 21^n$.**

Sol. $3^p 6^m 21^n = 2^m\ 3^{p+m+n}\ 7^n$

Therefore, the required number of proper divisors is equal to the number of selections of any number of 3's and 7's [$\because$ for odd divisors 2 must not be selected] and is given by
$(p + m + n + 1)(n + 1) - 1$.

Q. **Prove that number of ways in which N can be resolved as a product of 2 divisors which are relatively prime = 2^{n-1}. Where n is the number of prime involved in the prime factorisation of** $N = 2^6 \times 3^5 \times 5^4 \times 7^3 \times 11^2 \times 13^1$

Sol. $N = 2^6 \times 3^5 \times 5^4 \times 7^3 \times 11^2 \times 13^1$

a b c d e f a, b, c, d, e, f are relatively prime

Total ways $= {}^6C_0 + {}^6C_1 + {}^6C_2 + \frac{{}^6C_3}{2}$

${}^6C_0 \rightarrow 1 \times abcdef$

${}^6C_1 \rightarrow a \times bcdef, b \times acdef, c \times abdef, d \times abcef, e \times abcdf, f \times abcde,$

${}^6C_2 \rightarrow ab \times cdef, ae \times bcdf, \ldots$

$\frac{{}^6C_3}{2} \rightarrow$ 2 for equal size group

Total ways $= {}^6C_0 + {}^6C_1 + {}^6C_2 + \frac{{}^6C_3}{2} = 2^{6-1} = 32$

Similarly for n prime factorisation total ways $= 2^{n-1}$

Q. **Find the number of divisors of the number $N = 2^3 . 3^5 . 5^7 . 7^9$ which are perfect squares.**

Sol. Since the divisor is a perfect square, each prime factor must occur even number of times.
2 can be taken in 2 ways (2^0 or 2^2)
3 can be taken in 3 ways (3^0 or 3^2 or 3^4)
Similarly, 5 can be taken in 4 ways (5^0 or 5^2 or 5^4 or 5^6) and 7 can be taken in 5 ways (7^0 or 7^2 or 7^4 or 7^6 to 7^8)
Hence, total divisors which are perfect square $= 2 \times 3 \times 4 \times 5 = 120$

Q. **Find the number of ways in which the number 94864 can be resolved as a product of two factors.**

Sol. $94864 = 2^4 \times 7^2 \times 11^2$ (perfect square)
Hence, the number of ways is

$$\frac{1}{2}[(4+1)(2+1)(2+1)+1] = 23$$

Q. **Find the number of ways in which the number 300300 can be split into two factors which are relatively prime.**

Sol. $300300 = 2^2 \times 3^1 \times 5^2 \times 7^1 \times 11^1 \times 13^1$
Now we have to make factors which are relative prime.
Each of 2^2, 3^1, 5^2, 7^1, 11^1, 13^1 should be considered single factor.
So, number of divisors = $(1+1)(1+1)(1+1)(1+1)(1+1)(1+1) = 2^6 = 64$
Number of ways of splitting into 2 factors = $\frac{64}{2} = 32$

Q. **Let N be the number of ordered pairs of non-empty sets A and B that have the following properties:**
(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
(b) $A \cap B = \phi$
(c) The number of elements of A is not an element of A
(d) The number of elements of B is not an element of B
Find N.

Sol. **Explanation :**
(a) and (b) $\Rightarrow$ n(A) + n(B) = 10
(c) and (d) $\Rightarrow$ if A is the two elements set then B is an eight elements set, therefore 2 is not in A and must be in B and 8 is not in B and must be in A.
Also note that both set can not have equal number of elements because if A is a 5 element set then B will also be five element set and the elements can not be both in A and B.

No. of elements in A i.e. n(A)	n(B)	Number of ways
1 element set (say) {9}	{1...} 9 element set	$^8C_0$ ◆ group of 0 and 8

2 element set {8 ...}	{2...} 8 element set	$^8C_1$ ◆ group of 1 and 7
3 element set {7 ...}	{3...} 7 element set	$^8C_2$ ◆ group of 2 and 6
4 element set {6 ...}	{4...} 6 element set	$^8C_3$
5 element set	Not possible	
6 element set	4 element set	$^8C_5$
7 element set	3 element set	$^8C_6$
8 element set	2 element set	$^8C_7$
9 element set	1 element set	$^8C_8$

Total = $(^8C_0 + {}^8C_1 + {}^8C_2 + \ldots + {}^8C_8) - {}^8C_4 = 2^8 - 70 = 256 - 70 = 186$

Q. **The toal number of non-empty subsets S of the set A {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} such that, no two consecutive integers are in S and if, S have k elements, then no number less than k is in S.**

Sol. S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}
single element subset = 12
two element subset (1 cannot be taken) $^{10}C_2 = 45$
similarly 3 elements subset (1 and 2 cannot be taken)
3, 4, ... 11, 12
0 0 0

| × | × | × | × | × | × | × |

${}^{8}C_{3} = 56$

4 element set (1, 2, 3, rejected)

0 0 0 0

| × | × | × | × | × |

${}^{6}C_{4} = 15$

5 element (1, 2, 3, 4 rejected)

0 0 0 0 0

| × | × | × |

${}^{4}C_{5} = 0$ Not possible

Total = 12 + 45 + 56 + 15 = 68 + 60 = 128

SUMMATION OF NUMBERS (3 DIFFERENT WAYS)

(a) sum of all the numbers greater than 10000 formed by the digits 1, 3, 5, 7, 9 if no digit being repeated.

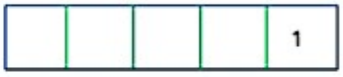
Method-1 :

All possible numbers = 5! = 120

If one occupies the units place then total numbers = 24

Hence 1 enjoys each place 24 times

Similarly, 1 enjoys each place 24 times



Sum due to 1 = $1 \times 24 (1 + 10 + 10^2 + 10^3 + 10^4)$

Similarly, sum due to the

digit 3 = $3 \times 24(1 + 10 + 10^2 + 10^3 + 10^4)$

⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮

Required total sum

$= 24(1 + 10 + 10^2 + 10^3 + 10^4)(1 + 3 + 5 + 7 + 9)$

.

Q. **Find the sum of all the numbers greater than 10000 formed by the digit 0, 1, 2, 4, 5, no digit being repeated.**

Sol. Using all the given digits we can form a five digit number except when zero is at first place.

So to find the sum of all the possible five digit number

= (Sum of all possible arrangement)

– (Sum of all the arrangements when zero is at first place)

⇒ 5 different digits can be arranged in 5! ways so each digit will appear at every place = $\frac{5!}{5}$ times i.e. 24 times

Sum of all digits at unit place = 24(0 + 1 + 2 + 4 + 5)
Sum of all digits at ten's place = 24(0 + 1 + 2 + 4 + 5)
⋮
⋮
Sum of all digits at 10000th place = 24(0 + 1 + 2 + 4 + 5)
In this way sum of all possible arrangement
= 24(0 + 1 + 2 + 4 + 5) [1 + 10 + 10^2 + 10^3 + 10^4]
when zero is at first place 4 digit number will be formed.
Each number will appear 6 times at every place.
Sum of all 4 digit number at unit place = 6(1 + 2 + 4 + 5)
Sum of all 4 digit number at ten's place = 6(1 + 2 + 4 + 5)
⋮
⋮
Hence sum of all four digit numbers = 6(1 + 2 + 4 + 5) (1 + 10 + 10^2 + 10^3)
Required sum = 24[0 + 1 + 2 + 3 + 4 + 5] [1 + 10 + 10^2 + 10^3 + 10^4]
−6(1 + 2 + 4 + 5) (1 + 10 + 10^2 + 10^3)

Q. **Find the sum of all the four digit number that can be formed with the digits 3, 2, 3, 4.**

Sol. The number of numbers having 2 in units place

$= \frac{3!}{2!} = 3$ [$\because$ the other three places are to be filled by 3, 3 and 4] Similarly the number of numbers having 4 in units places

$= \frac{3!}{2!} = 3$ [$\because$ the other three places are to be filled by 3, 3 and 2]

and the number of numbers having 3 in units places.
= 3! = 6 [$\because$ the other three places are to be filled by 2, 3 and 4]
Thus, sum of the digits occurring in the units place
=2 × 3 + 3 × 6 + 4 × 3 = 36
We can see that the given digits (3, 2, 3, 4) occur at the tens, hundreds and thousands place, the same number of times as they occur at the units place.
Hence, the required sum of the numbers formed.
= 36(1 + 10 + 100 + 1000) = 39996

Q. **Find the sum of the five digit numbers that can be formed using the digits 3, 4, 5, 6, 7 not using any digit more than once in any number.**

Sol. If 3 is placed at units place, the remaining 4 places can be filled in 4! = 24 ways
Thus, 3 occurs at unit places 24 times.
The other digits similarly, each occurs at the unit places 24 times.
Similarly, each of the digit occurs at the other places tens, hundreds and so on, 24 times.
Hence, the required sum, is

$=24(3 + 4 + 5 + 6 + 7)\ (10^0 + 10^1 + 10^2 + 10^3 + 10^4)$

$= 24 \times 25 \times 11111 = 6666600$

DISTRIBUTION OF ALIKE OBJECTS :

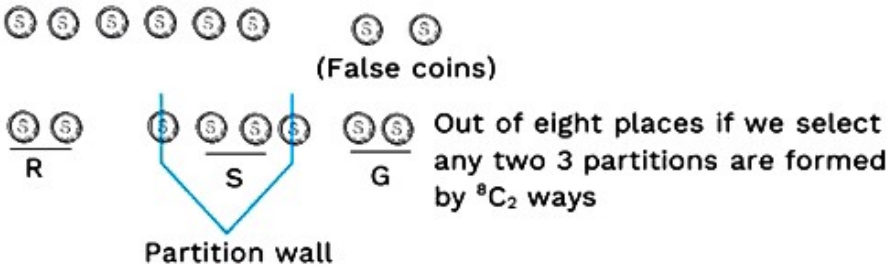
Type – 1 :

Total number of ways in which n identical coins can be distributed among p persons so that each person may get any number of coins is

$$^{n+p-1}C_{p-1} = \frac{(n+p-1)!}{(p-1)!(n)!}$$

Proof :

Let 6 identical coins can be distributed among 3 persons R|S|G



Out of eight places if we select any two 3 partitions are formed by $^8C_2$ ways

Q. **Find number of ways in which 30 mangoes can be distributed among 5 persons.**

Sol. $^{30+5-1}C_{5-1} = {}^{34}C_4$

Type – 2 :

Total number of ways in which n identical items can be distributed among p person such that each of them received at least one item $^{n-1}C_{p-1}$.

Q. **Find total number of ways distributing 7 identical computers to R|S|G, so that each receive at least one computer.**

Sol. ${}^{7-1}C_{3-1} = {}^{6}C_{2}$

Important Point

Number of different terms in a complete homogeneous expression of degree m in n variables is equivalent to distribution of m identical coins among n beggars.

If expression is $\left(x_1 + x_2 + x_3 \ldots + x_n\right)^m$, number of terms $= {}^{m+n-1}C_{n-1}$

Q. **Find number of different terms in expansion of $(x + y + z)^{10}$.**

Sol. ${}^{10+3-1}C_{3-1} = {}^{12}C_{2}$

Q. **Number of non-negative integral solution of the inequality $x + y + z + t \leq 30$**

Sol. $(x + y + z + t) + w = 30$

$\downarrow$

(False beggar)

Required condition is equivalent to giving 30 coins to 5 beggars

$= {}^{30+5-1}C_{5-1} = {}^{34}C_{4}$

Q. **Number of ways in which 16 identical toys are to be distributed among 3 children such that each child does not receive less than 3 toys will be**

(A) 96 **(B) 16** **(C) 36** **(D) 46**

Ans. **(C)**

Sol. Let x_1, x_2, x_3 be the number of toys received by the three children

Then, $x_1, x_2, x_3 \geq 3$ and $x_1 + x_2 + x_3 = 16$

Let $u_1 = x_1 - 3, u_2 = x_2 - 3$ and $u_3 = x_3 - 3$

Then, $u_1, u_2, u_3 \geq 0$ and $u_1 + u_2 + u_3 = 7$

Here, $n = 7$ and $r = 3$

$\therefore$ Number of ways $= {}^{n+r-1}C_{r-1} = {}^{9}C_{2} = 36$

Q. **Number of non-negative integral solutions of $x_1 + x_2 + x_3 + 4x_4 = 20$ will be**
(A) 496 **(B) 516** **(C) 536** **(D) 546**

Ans. **(C)**

Sol. Here, clearly $0 \le x_4 \le 5$, $x_1, x_2, x_3 \ge 0$ and $x_1 + x_2 + x_3 = 20 - 4x_4$

$\Rightarrow r = 3$ and $n = 20 - 4x_4$

If $x_4 = 0$, number of ways = ${}^{20+3-1}C_{3-1} = {}^{22}C_2$

If $x_4 = 1$, number of ways = ${}^{16+3-1}C_{3-1} = {}^{18}C_2$

Similarly, if $x_4 = 2, 3, 4, 5$, number of ways = ${}^{14}C_2, {}^{10}C_2, {}^{2}C_2$ respectively.

$\therefore$ Total number of ways

$= {}^{22}C_2 + {}^{18}C_2 + {}^{14}C_2 + {}^{10}C_2 + {}^{6}C_2 + {}^{2}C_2 = 536$

Q. **Number of ways 5 identical balls can be distributed into 3 different boxes so that no box remains empty will be**
(A) 1 **(B) 3** **(C) 6** **(D) 15**

Sol. **(C)**

The required number of ways ${}^{5-1}C_{3-1}$

$= {}^{4}C_2 = \frac{4.3}{1.2} = 6$

Q. **Find number of non-negative integral solutions of the equation $x + y + z = 10$.**

Sol. Here the number of solutions is equivalent to number of ways 10 identical objects can be distributed among three persons if empty groups are allowed that is ${}^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$.

Q. **Find number of positive integral solutions of the equation $x + y + z = 12$.**

Sol. Here the number of solutions is equivalent to the number of ways 12 identical objects can be distributed among three persons if empty groups are not allowed that is ${}^{12-1}C_{3-1} = {}^{11}C_2 = 55$

Q. **In an experiment, n six-faced normal dice are thrown. Find the number of sets of observations which are indistinguishable among themselves.**

Sol. Let number i be obtained x_i number of times.
So, we have $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = n$, where $x_i \geq 0$.
Hence, required number
= Number of non-negative integral solutions of above equation
$= {}^{n+6-1}C_{6-1}$
$= {}^{n+5}C_5$

Q. **Total number of positive integral solution in the form of (x, y, z) triplets such that this triplet satisfies the equation xyz = 24.**

Sol. Prime factorisation of $24 = 2^3 \times 3$
Suppose we have three boxes x, y, z, '3' can be put in any of the three boxes.
Also, 2, 2, 2 can be distributed in the three boxes in ${}^{3+3-1}C_{3-1} = {}^5C_2$ ways.
Hence, the total number of positive integral solutions is equal to the number of distributions which is given by $3 \times {}^5C_2 = 30$.

Q. **Consider the equation $\frac{2}{x}+\frac{5}{y}=\frac{1}{3}$, where x, y $\in$ N. Find the number of solutions of the equation.**

Sol. Given equation is $\frac{2}{x}+\frac{5}{y}=\frac{1}{3}$

or $6y + 15x = xy$
or $(x - 6)(y - 15) = 90 = 2 \times 3^2 \times 5$
Consider (x – 6) and (y – 15) as boxes.
Number of solutions is same as number of ways 2, 3, 3, 5 can be distributed in these boxes.
Number of ways 2 can be distributed = 2
Number of ways 5 can be distributed = 2
Number of ways (3, 3) can be distributed = 3
$\therefore$ Total number of solutions $= 2 \times 2 \times 3 = 12$

Q. **Number of ways in which 10 persons can take seat in a row of 24 seats so that no two persons sit on two consecutive seats?**

Sol. From the figure we can clearly see that x_0 is the number of vacant seats before first person and so on.



Clearly, the total number of vacant seats is 24 – 10 = 14.

$\therefore \quad x_0 + x_1 + x_2 + ... + x_9 + x_{10} = 14$

where $x_0 \geq 0, x_1 \geq 1, x_2 \geq 1, x_3 \geq 1, ..., x_9 \geq 1, x_{10} \geq 0$.

Let $x_0 = y_0, x_1 = y_1 + 1, x_2 = y_2 + 1, ..., x_9 = y_9 + 1, x_{10} = y_{10}$.

Then, the equations becomes

$y_0 + (y_1 + 1) + (y_2 + 1) + ... + (y_9 + 1) + y_{10} = 14$

or $y_0 + y_1 + y_2 + ... + y_9 + y_{10} = 14 - 9 = 5$

$\therefore$ Number of non-negative integral solutions of the above equation is

$^{5+11-1}C_{11-1} = {}^{15}C_{10}$.

But 10 persons can arrange among themselves in 10! Ways. Hence, the required number of ways is

$$^{15}C_{10} \times 10! = \frac{15!}{10!5!} \times 10! = \frac{15!}{5!}.$$

MAXIMISE ^nC_r

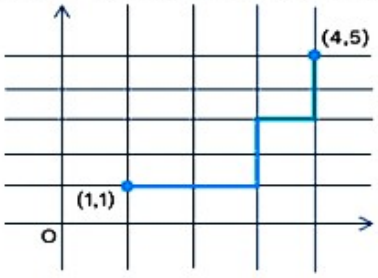
^nC_r is maximum at $\begin{cases} r = \frac{n}{2} & \text{if } n \text{ is even} \\ r = \frac{n-1}{2} \text{ or } \frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases}$

e.g. $^{15}C_r$ is maximum when r = 7 or 8, $^{12}C_r$ is maximum when r = 6

GRID PROBLEM :

Q. **Suppose we have given a graph paper, in which an ant is on point (1, 1), then find the number of ways in which that ant can travel to (4, 5) using the shortest path .**

Sol. For shortest path,ant has to travel 3 horizontal (H) and 4 vertical (V) paths Hence required number of path = $\frac{7!}{4!3!} = {}^7C_3$



Note :

If there are n vertical and m horizontal lines then there will be (n–1) horizontal and (m–1) vertical paths.

DISTRIBUTION OF DISTINCT OBJECTS IF EMPTY GROUPS ARE NOT ALLOWED

Let us establish that the formula for number of ways of distribution of n distinct objects among r persons if each gets at least one object is

$r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - {}^rC_3(r-3)^n + \ldots + (-1)^{r-1}\,{}^rC_{r-1}\,1$

Let A_i denotes the set of distribution of objects if i^{th} person gets no object and the other gets any number of objects.

$\therefore\ n(A_i) = (r-1)^n$ [as n objects can be distributed among (r – 1) persons in $(r-1)^n$ ways]

Then $n(A_i \cap A_j)$ represents number of distribution ways in which person i and j gets no object.

$\therefore\quad n(A_i \cap A_j) = (r-2)^n$

Similarly, $n(A_i \cap A_j \cap A_k) = (r-3)^n$

So, required number of ways

$= n(A_1' \cap A_2' \cap \ldots \cap A_r')$

$= r^n - n(A_1 \cup A_2 \cup \ldots \cup A_r)$

$$= r^n - \left[\sum_{i=1}^{n} n(A_i) - \sum_{1\le i<j\le n} n(A_i \cap A_j) + \sum_{1\le i<j<k\le n} n(A_i \cap A_j \cap A_k) + \ldots\right]$$

$= r^n - [{}^rC_1(r-1)^n - {}^rC_2(r-2)^n + {}^rC_3(r-3)^n - \ldots]$

$= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - {}^rC_3(r-3)^n + \ldots + (-1)^{r-1}\,{}^rC_{r-1}\,1$

Q. **Find the number of ways in which 5 distinct balls can be distributed in three different boxes if no box remains empty.**

or

If n(A) = 5 and n(B) = 3, then find the number of onto functions for A to B.

Sol. Using above formula, the number of ways is

$3^5 - {}^3C_1 \times (3-1)^5 + {}^3C_2(3-2)^5 = 243 - 96 + 3 = 150.$

DERANGEMENT

We have n different letters and n different envelopes, the number of ways by which we can put letters in the envelopes such that each envelope has 1 letter and no letters is in their corresponding envelope. This condition can also be written as "Number of ways so that n different things can deranged from their original places and is represented by Dn :

Where ,

$$D_n = n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\ldots+(-1)^n\frac{1}{n!}\right)$$

Thus, we have

$D_0 = 0! = 1$

$D_1 = 1! - {}^1C_1 D_0 = 0$

$D_2 = 2! - {}^2C_1 D_1 - {}^2C_2 D_0 = 1$

$D_3 = 3! - {}^3C_1 D_2 - {}^3C_2 D_1 - {}^3C_3 D_0 = 2$

$D_4 = 4! - {}^4C_1 D_3 - {}^4C_2 D_2 - {}^4C_3 D_1 - {}^4C_4 D_0 = 9$

$D_5 = 5! - {}^5C_1 D_4 - {}^5C_2 D_3 - {}^5C_3 D_2 - {}^5C_4 D_1 - {}^5C_5 D_0 = 44$

and so on

Q. **There are 5 boxes of 5 different colours. Also, there are 5 balls of same colours as those of the boxes. The number of ways in which we can place 5 balls in 5 boxes such that at most 2 balls goes to the wrong boxes, is**
(A) 9 **(B) 10** **(C) 11** **(D) None of these**

Sol. **(D)**

At most 2 balls goes to the wrong boxes = 2 balls to the wrong boxes and 3 to the right boxes + 1 ball to the wrong box and 4 balls to the right boxes + 0 ball to wrong box and 5 balls to the right boxes

$$= 2!\left(1-\frac{1}{1!}+\frac{1}{2!}\right)\times {}^5C_3 + 0 + 1$$

$= 1 \times 10 + 0 + 1 = 11$

[(not possible to place 1 ball in wrong box) + 1 (all balls goes to right boxes)]

Q. **Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is**

(A) 264 **(B) 265** **(C) 53** **(D) 67**

Sol. **(C)**

Using the principle of inclusion and exclusion, we have the number of ways in which card number 1 be placed in envelope number 2

$= 5! - \left\{ {}^4C_1 \cdot 4! - {}^4C_2 \cdot 3! + {}^4C_3 \cdot 2! - {}^4C_4 \cdot 1! \right\}$

$= 120 - \{96 - 36 + 8 - 1\} = 53$

Q. **Find the number of ways of distributing 5 different balls in three boxes of different size so that no box is empty and each box being large enough to accommodate all the five balls.**

(A) 100 **(B) 120** **(C) 140** **(D) 150**

Sol. **(A)**

Using the fact:

Number of ways of dividing n non-identical objects in r groups such that each group get at least one object i.e. empty groups are not allowed

$= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - {}^rC_3(r-3)^n + \ldots + (-1)^{r-1} {}^rC_{r-1} 1^n$

$= 3^5 - {}^3C_1 \cdot 2^5 + {}^3C_2(1)^5$ (Putting n = 5, r = 3)

$= 243 - 3 \times 32 + 3 = 150$

$D(3) = 3!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right) = 2$

Therefore, total number of required ways $= {}^6C_3 \times 3! \times 2 = 240$